

Rational minimax approximation of matrix-valued functions

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Abstract

In this paper, we present a rigorous framework for rational minimax approximation of matrix-valued functions that generalizes classical scalar approximation theory. Given sampled data $\{(x_\ell, F(x_\ell))\}_{\ell=1}^m$ where $F : \mathbb{C} \rightarrow \mathbb{C}^{s \times t}$ is a matrix-valued function, we study the problem of finding a matrix-valued rational approximant $R(x) = P(x)/q(x)$ (with $P : \mathbb{C} \rightarrow \mathbb{C}^{s \times t}$ a matrix-valued polynomial and $q(x)$ a nonzero scalar polynomial of prescribed degrees) that minimizes the worst-case Frobenius norm error over the given nodes:

$$\inf_{R(x)=P(x)/q(x)} \max_{1 \leq \ell \leq m} \|F(x_\ell) - R(x_\ell)\|_F.$$

By reformulating this min-max optimization problem through duality, we derive a maximization dual problem over the probability simplex. We analyze weak and strong duality properties and establish a sufficient condition ensuring that the solution of the dual problem yields the minimax approximant $R(x)$. For numerical implementation, we propose an efficient method (**m-d-Lawson**) to solve the dual problem, generalizing Lawson's iteration to matrix-valued functions. Convergence analysis of **m-d-Lawson** is established. Numerical experiments are conducted and compared to state-of-the-art approaches, demonstrating its efficiency as a novel computational framework for matrix-valued rational approximation.

Keywords: minimax problem, duality, rational approximation for matrix-valued functions, Lawson's iteration, minimax approximation

AMS subject classifications: 49J35 , 49K35 , 90C47 , 90C46 , 41A50 , 41A20 , 65D15

1 Introduction

Let $F : \mathbb{C} \rightarrow \mathbb{C}^{s \times t}$ be a matrix-valued function defined on a subset of Ω in the complex plane. For a given set of distinct nodes $\mathcal{X} = \{x_\ell\}_{\ell=1}^m$ in Ω , suppose that we have the discrete sampled data $\{(x_\ell, F(x_\ell))\}_{\ell=1}^m$ with

$$F(x_\ell) = \begin{bmatrix} f_{11}(x_\ell) & \cdots & f_{1t}(x_\ell) \\ \vdots & \ddots & \vdots \\ f_{s1}(x_\ell) & \cdots & f_{st}(x_\ell) \end{bmatrix} \in \mathbb{C}^{s \times t}. \quad (1.1)$$

Let $\|F(x_\ell)\|_F$ be the Frobenius norm of $F(x_\ell)$ and $\mathbb{P}_n$ be the set of complex polynomials with degree less than or equal to n . In this paper, we consider the following discrete rational minimax approximation for the matrix-valued function $F(x)$:

$$\eta_\infty := \inf_{R \in \mathcal{R}_{(s,t)}} \max_{1 \leq \ell \leq m} \|F(x_\ell) - R(x_\ell)\|_F^2, \quad (1.2)$$

where

$$\mathcal{R}_{(s,t)} := \left\{ R(x) = \begin{bmatrix} r_{11}(x) & \cdots & r_{1t}(x) \\ \vdots & \ddots & \vdots \\ r_{s1}(x) & \cdots & r_{st}(x) \end{bmatrix} \mid r_{ij}(x) = \frac{p_{ij}(x)}{q(x)}, p_{ij} \in \mathbb{P}_{n_{ij}}, 0 \neq q \in \mathbb{P}_d \right\}, \quad (1.3)$$

$n_{ij} \geq 0$ and $d \geq 0$ are prescribed integers. The (i, j) th function $\frac{p_{ij}(x)}{q(x)}$ of $R(x)$ is said to be a rational function of type (n_{ij}, d) where $p_{ij} \in \mathbb{P}_{n_{ij}}$ is the (i, j) entry of a matrix-valued polynomial $P : \mathbb{C} \rightarrow \mathbb{C}^{s \times t}$. Moreover, without loss of generality, we assume throughout the paper that

$$m \geq \max_{1 \leq i \leq s, 1 \leq j \leq t} \{n_{ij} + d + 2\}, \quad (1.4)$$

since otherwise, $R \in \mathcal{R}_{(s,t)}$ simplifies to an interpolation problem for $\{(x_\ell, F(x_\ell))\}_{\ell=1}^m$.

It is clear that (1.2) generalizes the classical discrete rational minimax approximation for complex-valued functions (e.g., [13, 21, 34, 36, 37, 40, 42, 43, 44, 48, 50, 51]), reducing to the classical case when $s = t = 1$. For clarity, we hereafter refer to (1.2) as the matrix-valued (or block) discrete rational minimax approximation. Following the setting of matrix-valued rational approximation [4, 5, 16, 19, 28, 52], we enforce a common denominator $q \in \mathbb{P}_d$ for all entries $r_{ij}(x)$ in (1.2).

Notably, the matrix-valued rational minimax approximation arises in diverse applications, including the linear time-invariant system (e.g., [5, 16]), frequency-domain multiport rational modeling (see e.g., [10, 16, 18, 19, 20]), computer-aided designing of the microwave duplexer [38, 39, 52] and nonlinear eigenvalue problems [22, 23, 28, 35, 46]. In applications such as nonlinear eigenvalue problems, minimax rational approximants $R(x)$ are sought for matrix-valued functions $F(x)$ over a continuum compact Ω bounded by a simple Jordan curve $\partial\Omega$ (see [46] for the applications of (1.2) to the nonlinear eigenvalue problems). When each entry $f_{ij}(x)$ is analytic in its interior and continuous on the boundary $\partial\Omega$, the maximum Frobenius norm principle holds:

$$\max_{x \in \Omega} \|F(x) - R(x)\|_F^2$$

is attained on $\partial\Omega$ for any $R(x)$ with poles outside Ω (see Theorem 1 in [9]). Consequently, solving problem (1.2) with carefully chosen boundary nodes $\mathcal{X} \subset \partial\Omega$ provides an efficient numerical approach for computing minimax rational approximants on Ω .

In the literature, several numerical methods exist for the discrete rational approximation of matrix-valued functions. Particularly, Gosea and Güttel [16] investigated a range of efficient algorithms, including the rational Krylov fitting (RKFIT) method [5], set-valued AAA [28], Loewner framework interpolation approach [2, 17, 29], and the vector fitting (VF) approach in [11, 19]; additionally, they introduced a block-AAA extension of the AAA algorithm [30] and conducted extensive numerical experiments to evaluate these methods. To approximate three rational functions emerging in microwave duplexer filter design [38, 39], a recent study [52] proposes a vector-valued AAA-type approach (v-AAA-Lawson), building upon the AAA algorithm [30] and AAA-Lawson method [14]; v-AAA-Lawson is also applicable to general matrix-valued rational approximations.

However, we observe that most existing algorithms lack a rigorously defined optimization objective, leading to approximations with no guaranteed optimality. As an extension of least squares, the absolute root mean squared error (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{m} \sum_{\ell=1}^m \sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_\ell) - r_{ij}(x_\ell)|^2} = \sqrt{\frac{1}{m} \sum_{\ell=1}^m \|F(x_\ell) - R(x_\ell)\|_F^2} \quad (1.5)$$

or the relative one is commonly adopted as an accuracy measure for a computed approximation $R : \mathbb{C} \rightarrow \mathbb{C}^{s \times t}$ [16]. However, there is no general algorithm that guarantees convergence to the globally optimal solution with minimal RMSE (see [5, A.5]).

Contributions. In this paper, with a rigorous and clear objective (1.2) in the minimax sense, we study the problem of finding the associated matrix-valued discrete rational minimax approximant. Exploring recent advances in duality-based treatments for the minimax approximation [45, 48, 50], we develop a numerical method to solve (1.2). Our key contributions are as follows.

- 1). Theoretical contributions:

- reformulation of the minimax problem (1.2) into a single-level minimization via linearization, along with a detailed analysis of their theoretical connections;
- derivation of a maximization dual problem of the linearization through Wolfe duality. We establish weak and strong duality between the linearized problem and its dual, and provide a sufficient condition (Theorem 4.1) ensuring that the solution of the dual problem yields the best approximant for (1.2).

2). Computational contributions:

- an extension, namely **m-d-Lawson**, of the **d-Lawson** iteration [50] to solve (1.2) via the dual problem;
- convergence analysis of **m-d-Lawson** is established;
- numerical validation demonstrating the efficiency of **m-d-Lawson** against state-of-the-art approaches, qualifying it as a novel computational framework for matrix-valued rational approximation.

Paper organization. The structure of the remaining paper is summarized below. Section 2 introduces a linearization of (1.2) and discusses its relation with (1.2); the duality theory is applied to this linearization, yielding a maximization dual problem. In Section 3, we focus on the implementation details of computing the dual function at a given dual variable; particularly, the Vandermonde with Arnoldi process [7] is introduced in this computation step. Section 4 is devoted to strong duality between the linearization of (1.2) and the dual problem. Under strong duality, in Section 5, we develop a complementary slackness property at the maximizer of the dual problem and discuss its connection with the *extreme points* (aka *reference points*) of the best approximant of (1.2). The new method **m-d-Lawson** is proposed in Section 6, which generalizes the classical Lawson’s iteration [26] to the matrix-valued functions. The convergence analysis of **m-d-Lawson** is presented in Section 7. Section 8 provides numerical experiments comparing **m-d-Lawson** with state-of-the-art methods for matrix-valued rational approximation, followed by detailed performance analysis. Finally, Section 9 summarizes the key findings and conclusions. The Appendix section collects several proofs of the paper.

Notation. Throughout this paper, we employ the following notation:

- $\mathbb{C}^{n \times m}$ (resp. $\mathbb{R}^{n \times m}$) is the set of all $n \times m$ complex (resp. real) matrices and $\bullet^H$ (resp. $\bullet^T$) represents the conjugate transpose (resp. transpose).
- All real or complex numbers (or functions) are denoted by Latin (e.g., a) or Greek letters (e.g., α). Column vectors are represented by boldface lowercase Latin (e.g., $\mathbf{a}$) or Greek letters (e.g., $\boldsymbol{\alpha}$), where boldface lowercase symbols with subscripts (e.g., $\mathbf{p}_{ij}$ and $\boldsymbol{\tau}_i$) also denote column vectors. Matrices are denoted by boldface or non-boldface uppercase Latin (e.g., A and $\mathbf{A}$) or Greek letters (e.g., $\boldsymbol{\Phi}$); similarly, uppercase letters with subscripts (e.g., F_{ij} and $\boldsymbol{\Psi}_{ij}$) also represent matrices. For simplicity, we sometimes express column vectors and matrices in terms of their elements, as $\mathbf{a} = [a_1, \dots, a_n]^T$ and $A = [a_{ij}]$, respectively.
- $I_n \equiv [\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n] \in \mathbb{R}^{n \times n}$ represents the n -by- n identity matrix, where $\mathbf{e}_i$ is its i th column with $1 \leq i \leq n$. The vector $\mathbf{1}_g = [1, \dots, 1]^T \in \mathbb{R}^g$ is the g -dimensional all-ones column vector.

- $\mathbf{i} = \sqrt{-1}$ represents the imaginary unit, and for $\mu \in \mathbb{C}$, we write $\mu = \mu^{\mathbf{r}} + \mathbf{i}\mu^{\mathbf{i}}$ and $\bar{\mu} = \mu^{\mathbf{r}} - \mathbf{i}\mu^{\mathbf{i}}$ where $\text{Re}(\mu) = \mu^{\mathbf{r}} \in \mathbb{R}$, $\text{Im}(\mu) = \mu^{\mathbf{i}} \in \mathbb{R}$ and $|\mu| = \sqrt{(\mu^{\mathbf{r}})^2 + (\mu^{\mathbf{i}})^2}$.
- $\text{diag}(\mathbf{x}) = \text{diag}(x_1, \dots, x_n)$ denotes the diagonal matrix associated with the vector $\mathbf{x} \in \mathbb{C}^n$, and $\mathbf{x} \cdot \mathbf{y} = \frac{\mathbf{x}}{\mathbf{y}} = [x_1/y_1, \dots, x_n/y_n]$ for two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$ with $y_j \neq 0$, $1 \leq j \leq n$.
- $\text{span}(A)$ is a subspace spanned by columns of A ; the k th Krylov subspace generated by a matrix A on $\mathbf{b}$ is denoted by $\mathcal{K}_k(A, \mathbf{b}) = \text{span}(\mathbf{b}, A\mathbf{b}, \dots, A^{k-1}\mathbf{b})$.

2 The dual problem

2.1 Representation of rational functions

In order to represent a rational function $r_{ij}(x) = \frac{p_{ij}(x)}{q(x)}$, we can choose basis functions $\{\psi_0(x), \dots, \psi_{n_{ij}}(x)\}$ and $\{\phi_0(x), \dots, \phi_d(x)\}$ to parameterize

$$r_{ij}(x) = \frac{p_{ij}(x)}{q(x)} = \frac{[\psi_0(x), \dots, \psi_{n_{ij}}(x)]\mathbf{a}_{ij}}{[\phi_0(x), \dots, \phi_d(x)]\mathbf{b}}, \quad \text{for some } \mathbf{a}_{ij} \in \mathbb{C}^{n_{ij}+1}, \mathbf{b} \in \mathbb{C}^{d+1}.$$

In principle, $\{\psi_0(x), \dots, \psi_{n_{ij}}(x)\}$ and $\{\phi_0(x), \dots, \phi_d(x)\}$ can be any basis functions for $\mathbb{P}_{n_{ij}}$ and $\mathbb{P}_d$, respectively. Besides the monomial bases, the barycentric representation for the rational functions [14, 16, 30, 51] can also be a choice; a recent work in [1] also generalizes the barycentric form to the multivariate rational approximations arising from linear parametric systems. While any parameterization that is able to represent the rational functions is mathematically equivalent, numerically, particular implementations and techniques should be concerned to deal with the numerical stability. In this paper, we will mainly use the monomial basis $\psi_i(x) = \phi_i(x) = x^i$ and rely on the Vandermonde with Arnoldi process [7, 25, 49] to handle the ill-conditioned problems inherited from the Vandermonde matrix at nodes $\mathcal{X} = \{x_\ell\}_{\ell=1}^m$; in particular, for $p_{ij}(x)$ in the rational function $r_{ij}(x) = \frac{p_{ij}(x)}{q(x)}$, we denote by

$$\mathbf{\Psi}_{ij} = \mathbf{\Psi}_{ij}(x_1, \dots, x_m; n_{ij}) := \begin{bmatrix} \psi_0(x_1) & \psi_1(x_1) & \cdots & \psi_{n_{ij}}(x_1) \\ \psi_0(x_2) & \psi_1(x_2) & \cdots & \psi_{n_{ij}}(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_0(x_m) & \psi_1(x_m) & \cdots & \psi_{n_{ij}}(x_m) \end{bmatrix} \in \mathbb{C}^{m \times (n_{ij}+1)} \quad (2.1)$$

the associated Vandermonde matrix; therefore,

$$\mathbf{p}_{ij} := [p_{ij}(x_1), \dots, p_{ij}(x_m)]^T = \mathbf{\Psi}_{ij}\mathbf{a}_{ij} \in \mathbb{C}^m. \quad (2.2)$$

Similarly, we denote by $\mathbf{\Phi} = \mathbf{\Phi}(x_1, \dots, x_m; d) \in \mathbb{C}^{m \times (d+1)}$ the associated Vandermonde matrix for the denominator $q(x)$ at $\mathcal{X}$, whose (i, j) th entry is $\phi_{j-1}(x_i)$; thus

$$\mathbf{q} := [q(x_1), \dots, q(x_m)]^T = \mathbf{\Phi}\mathbf{b} \in \mathbb{C}^m. \quad (2.3)$$

In passing, due to the common denominator $q(x)$ shared by all $r_{i,j}(x)$, we remark that the entry functions in the solution $R(x)$ of (1.2), if it exists, are distinct from

those acquired independently through the rational minimax approximations of $f_{ij}(x)$ ($1 \leq i \leq s$ and $1 \leq j \leq t$) on $\mathcal{X}$. For the special matrix-valued polynomial case with $d = 0$, i.e., $q \equiv 1$, indeed, we have

$$\begin{aligned}\eta_\infty &= \inf_{\{p_{ij}\}} \max_{1 \leq \ell \leq m} \sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_\ell) - p_{ij}(x_\ell)|^2 \\ &\leq \inf_{\{p_{ij}\}} \sum_{i=1}^s \sum_{j=1}^t \max_{1 \leq \ell \leq m} |f_{ij}(x_\ell) - p_{ij}(x_\ell)|^2 \\ &= \sum_{i=1}^s \sum_{j=1}^t \inf_{p_{ij} \in \mathbb{P}_{n_{ij}}} \max_{1 \leq \ell \leq m} |f_{ij}(x_\ell) - p_{ij}(x_\ell)|^2,\end{aligned}$$

implying that (1.2) is better than independently approximating $f_{ij}(x)$ ($1 \leq i \leq s$, $1 \leq j \leq t$) over $\mathcal{X}$.

Given an $R(x) \in \mathcal{R}_{(s,t)}$, if the associated maximum approximation error

$$e(R) := \max_{x_\ell \in \mathcal{X}} \|F(x_\ell) - R(x_\ell)\|_F^2 \quad (2.4)$$

is finite, then we can assume, without loss of generality, that $q(x_j) \neq 0$, $\forall x_j \in \mathcal{X}$. To see this, note that if $q(z) = 0$ for some $z \in \mathcal{X}$, then z must be a zero for all the numerators $p_{ij}(x)$ with $1 \leq i \leq s$ and $1 \leq j \leq t$, because otherwise $e(R) = \infty$. This implies that zeros z of $q(x)$ in $\mathcal{X}$ can be canceled in all $p_{ij}(x)/q(x)$, and the values $r_{ij}(z)$ are well-defined through cancellation of these common zeros (i.e., this induces an equivalence in $\mathcal{R}_{(s,t)}$). Due to this fact, if a solution exists for (1.2), there is an $R \in \mathcal{R}_{(s,t)}$ such that $q(x_j) \neq 0$ for any $x_j \in \mathcal{X}$.

2.2 The dual problem and weak duality

Following [50], a key observation for addressing (1.2) is the following equivalence:

$$\max_{1 \leq \ell \leq m} \|F(x_\ell) - R(x_\ell)\|_F^2 \leq \eta \iff \|F(x_\ell) - R(x_\ell)\|_F^2 \leq \eta, \quad \forall 1 \leq \ell \leq m.$$

Providing $q(x_\ell) \neq 0$ ($\forall 1 \leq \ell \leq m$), the latter m constraints can further be represented as

$$\sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_\ell)q(x_\ell) - p_{ij}(x_\ell)|^2 \leq \eta |q(x_\ell)|^2, \quad 1 \leq \ell \leq m. \quad (2.5)$$

Thus, similar to the scalar version in [50], we can construct the following minimization

$$\begin{aligned}\eta_2 &:= \inf_{\eta \in \mathbb{R}, p_{ij} \in \mathbb{P}_{n_{ij}}, 0 \neq q \in \mathbb{P}_d} \eta \\ \text{s.t.}, & \sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_\ell)q(x_\ell) - p_{ij}(x_\ell)|^2 \leq \eta |q(x_\ell)|^2, \quad 1 \leq \ell \leq m.\end{aligned} \quad (2.6)$$

In Theorem 2.1, we prove that the infimum η_2 of (2.6) is indeed attainable, qualifying this linearization as a standard optimization. This is a parallel generalization of [48, Theorem 2.1] whose proof can be suitably modified to lead the following assertion and hence is provided in Appendix¹ A.

Theorem 2.1. *Given $m \geq \max_{1 \leq i \leq s, 1 \leq j \leq t} \{n_{ij} + d + 2\}$ distinct nodes $\mathcal{X} = \{x_\ell\}_{\ell=1}^m$ in $\mathbb{C}$, let η_2 be the infimum of (2.6). Then η_2 is attainable by certain polynomials $p_{ij} \in \mathbb{P}_{n_{ij}}$ ($1 \leq i \leq s, 1 \leq j \leq t$) and $0 \neq q \in \mathbb{P}_d$.*

Proof. See Appendix A. □

Following [48] again, we next provide a sufficient condition: $0 \leq d \leq \min_{1 \leq i \leq s, 1 \leq j \leq t} \{n_{ij}\}$ to ensure $\eta_\infty = \eta_2$. Note that such a sufficient condition includes both the matrix-valued polynomial case (i.e., $d = 0$) as well as the rational case with all rational r_{ij} of type (d, d) . The result, together with Theorem 2.1, paves the way to solve (1.2) through the linearization (2.6).

Theorem 2.2. *Given $m \geq \max_{1 \leq i \leq s, 1 \leq j \leq t} \{n_{ij} + d + 2\}$ distinct nodes $\mathcal{X} = \{x_j\}_{j=1}^m$ on $\Omega \subset \mathbb{C}$, let η_2 be the infimum of (2.6). If $0 \leq d \leq \min_{1 \leq i \leq s, 1 \leq j \leq t} \{n_{ij}\}$, then $\eta_2 = \eta_\infty$; furthermore, in this case, whenever (1.2) has a solution $R \in \mathcal{R}_{(s,t)}$ with $q(x_j) \neq 0$ ($\forall 1 \leq j \leq m$), the tuple $(\eta_\infty, \{p_{ij}\}, q)$ is a solution of (2.6).*

Proof. See Appendix B. □

Note that while (1.2) and its linearized counterpart (2.6) share the same infimum η_∞ , this value may not be attainable in (1.2)—a fundamental feature of rational minimax approximation. The following example extends [48, Example 2.1] to the matrix-valued setting, revealing the intrinsic distinction between (1.2) and (2.6).

Example 2.1. Consider $s = t = 2$, $n_{ij} = d = 1$, $\forall 1 \leq i, j \leq 2$ and $m = 4$ with $x_\ell = \frac{\ell-1}{4}$ ($1 \leq \ell \leq 4$). Let $F(x_1) = I_2$ and $F(x_\ell) = 0_{2 \times 2}$ for $2 \leq \ell \leq 4$. It is true that for (2.6), the infimum $\eta_2 = 0$ is attainable by $p_{ij}(x) \equiv 0$, $\forall i, j$ and $q(x) = x$. For (1.2), consider the sequence $R^{(k)}(x) = [r_{ij}^{(k)}(x)] \in \mathcal{R}_{(2,2)}$ with $r_{ij}^{(k)}(x) \equiv 0$ for $1 \leq i \neq j \leq 2$, while $r_{ij}^{(k)}(x) = \frac{1}{1+kx}$ for $1 \leq i = j \leq 2$. It holds that

$$\max_{1 \leq \ell \leq 4} \|F(x_\ell) - R^{(k)}(x_\ell)\|_F^2 = \frac{2}{(1 + \frac{k}{4})^2} \rightarrow 0, \quad \text{as } k \rightarrow \infty,$$

which implies $\eta_2 = \eta_\infty$. However, the infimum $\eta_\infty = 0$ is unattainable in (1.2) as any nonconstant type (1,1) rational function can only take each value once.

Theorem 2.1 reveals that (2.6) is a concrete minimization problem, for which a dual problem can be defined analogously to the scalar-valued case [6, 32]. To this end, we should note that each constraint in (2.6) is homogeneous in $(\{p_{ij}\}, q)$: for any $\lambda \neq 0$, replacing $(\{p_{ij}\}, q)$ by $(\{\lambda p_{ij}\}, \lambda q)$ leaves the constraint unchanged. Because of this scaling invariance, instead of the classical Lagrangian dual theory, the Wolfe dual [32, Section 12.9] should be applied. In particular, writing $p_{ij}(x_\ell)$ (resp., $q(x_\ell)$), that is,

¹Given the analogy with the scalar-valued minimax theory of [50, 48], we organize proofs of the paper as follows: results requiring only slight adaptations from the scalar case appear in the Appendix, while those needing significant modifications for the matrix-valued setting are proved in the main text.

the value of $p_{ij}(x)$ (resp., $q(x)$) at x_ℓ , as $p_{ij}(x_\ell; \mathbf{a}_{ij})$ (resp., $q(x_\ell; \mathbf{b})$), and introducing the dual variable $\mathbf{w}$, it leads to the following dual function over $\mathbf{w} \geq \mathbf{0}$:

$$\inf_{\substack{\mathbf{a}_{ij} \in \mathbb{C}^{n_{ij}+1}, \mathbf{b} \in \mathbb{C}^{d+1} \\ \sum_{\ell=1}^m w_\ell |q(x_\ell; \mathbf{b})|^2 = 1}} \sum_{\ell=1}^m w_\ell \left(\sum_{i=1}^s \sum_{j=1}^t \left| f_{ij}(x_\ell) q(x_\ell; \mathbf{b}) - p_{ij}(x_\ell; \mathbf{a}_{ij}) \right|^2 \right). \quad (2.7)$$

To simplify our presentation, associated with (2.7), let

$$n = \sum_{i=1}^s \sum_{j=1}^t (n_{ij} + 1), \quad g = st, \quad F_{ij} = \text{diag}(f_{ij}(x_1), \dots, f_{ij}(x_m)) \in \mathbb{C}^{m \times m}, \quad (2.8)$$

$$\mathbf{F} = \begin{bmatrix} F_{11} \\ \vdots \\ F_{s1} \\ F_{12} \\ \vdots \\ F_{st} \end{bmatrix} \in \mathbb{C}^{gm \times m}, \quad \mathbf{\Theta} = \begin{bmatrix} \Psi_{11} & & & & \\ & \ddots & & & \\ & & \Psi_{s1} & & \\ & & & \Psi_{21} & \\ & & & & \ddots \\ & & & & & \Psi_{st} \end{bmatrix} \in \mathbb{C}^{gm \times n}, \quad (2.9)$$

where the row blocks of $\mathbf{F}$ and the diagonal blocks of $\mathbf{\Theta}$ are ordered according to the rule of vectorizing the matrix column-wise (i.e., stacking the $(i+1)$ th column below the i th). Similarly, we stack vectors $\mathbf{a}_{ij} \in \mathbb{C}^{n_{ij}+1}$ and $\mathbf{p}_{ij} \in \mathbb{C}^m$ associated with the numerators p_{ij} to have

$$\mathbf{a} = [\mathbf{a}_{11}^T, \dots, \mathbf{a}_{s1}^T, \mathbf{a}_{12}^T, \dots, \mathbf{a}_{st}^T]^T \in \mathbb{C}^n, \quad \mathbf{p} = \mathbf{\Theta} \mathbf{a} = [\mathbf{p}_{11}^T, \dots, \mathbf{p}_{s1}^T, \mathbf{p}_{12}^T, \dots, \mathbf{p}_{st}^T]^T \in \mathbb{C}^{mg} \quad (2.10)$$

and define

$$\mathbf{W}_\otimes = I_g \otimes W \in \mathbb{C}^{gm \times gm}, \quad W = \text{diag}(\mathbf{w}) \in \mathbb{C}^{m \times m}, \quad (2.11)$$

where $\otimes$ stands for the Kronecker product. Consequently, we rewrite (2.7) compactly as

$$\inf_{\substack{\mathbf{a} \in \mathbb{C}^n, \mathbf{b} \in \mathbb{C}^{d+1} \\ \|\sqrt{W} \mathbf{\Phi} \mathbf{b}\|_2 = 1}} \left\| \sqrt{W_\otimes} (\mathbf{F} \mathbf{\Phi} \mathbf{b} - \mathbf{\Theta} \mathbf{a}) \right\|_2^2. \quad (2.12)$$

An immediate duality conclusion is the so-called *weak duality*, which asserts that the infimum of (2.12) at any $\mathbf{w} \geq \mathbf{0}$ is a lower bound for η_∞ . The following theorem as well as its proof can be viewed as the extension of [50, Theorem 2.2] to the matrix-valued rational minimax approximation (1.2).

Theorem 2.3. *Let η_∞ be the supremum of (1.2). Then we have weak duality:*

$$\sup_{\mathbf{w} \geq \mathbf{0}} d(\mathbf{w}) = \max_{\mathbf{w} \in \mathcal{S}} d(\mathbf{w}) \leq \eta_\infty, \quad (2.13)$$

where, using the notation in (2.8), (2.9), (2.10) and (2.11), the dual function $d(\mathbf{w})$ is

$$\begin{aligned} d(\mathbf{w}) &:= \min_{\substack{\mathbf{a}_{ij} \in \mathbb{C}^{n_{ij}+1}, \mathbf{b} \in \mathbb{C}^{d+1} \\ \sum_{\ell=1}^m w_\ell |q(x_\ell; \mathbf{b})|^2 = 1}} \sum_{\ell=1}^m w_\ell \left(\sum_{i=1}^s \sum_{j=1}^t \left| f_{ij}(x_\ell) q(x_\ell; \mathbf{b}) - p_{ij}(x_\ell; \mathbf{a}_{ij}) \right|^2 \right) \\ &= \min_{\substack{\mathbf{a} \in \mathbb{C}^n, \mathbf{b} \in \mathbb{C}^{d+1} \\ \|\sqrt{W}\Phi\mathbf{b}\|_2=1}} \left\| \sqrt{W_\otimes}(\mathbf{F}\Phi\mathbf{b} - \Theta\mathbf{a}) \right\|_2^2, \end{aligned} \quad (2.14)$$

and $\mathcal{S}$ is the probability simplex defined by

$$\mathcal{S} := \{\mathbf{w} = [w_1, \dots, w_m]^T \in \mathbb{R}^m : \mathbf{w} \geq 0 \text{ and } \mathbf{w}^T \mathbf{e} = 1\}, \quad \mathbf{e} = [1, 1, \dots, 1]^T. \quad (2.15)$$

Proof. See Appendix C. $\square$

Weak duality suggests the following max-min type dual problem

$$\max_{\mathbf{w} \in \mathcal{S}} d(\mathbf{w}) \quad (2.16)$$

to solve the original matrix-valued rational minimax approximation (1.2). In this framework, the success in computing the minimax solution $R^* \in \mathcal{R}_{(s,t)}$ of (1.2), if it exists, is crucially dependent on the following condition

$$d^* := \max_{\mathbf{w} \in \mathcal{S}} d(\mathbf{w}) = \eta_\infty. \quad (2.17)$$

The condition (2.17) implies that there is no gap between the maximum of the dual problem with η_∞ , and is known as *strong duality* (see e.g., [6, Chapter 5.2]) in optimization.

- (1) In polynomial minimax approximation for a complex-valued function f (i.e., $s = t = 1$, $d = 0$), strong duality is guaranteed (see [45, Theorem 2.1]). In this case, as $q \in \mathbb{P}_0$ and $\sum_{\ell=1}^m w_\ell |q(x_\ell)|^2 = 1$, $|q| \equiv 1$; thus $d(\mathbf{w})$ in (2.14) becomes

$$d(\mathbf{w}) = \min_{\mathbf{a} \in \mathbb{C}^n} \left\| \sqrt{W}(q\mathbf{F}\mathbf{e} - \Theta\mathbf{a}) \right\|_2^2 \stackrel{\tilde{\mathbf{a}} = \mathbf{a}/q}{=} \min_{\tilde{\mathbf{a}} \in \mathbb{C}^n} \left\| \sqrt{W}(\mathbf{f} - \Theta\tilde{\mathbf{a}}) \right\|_2^2,$$

where $\mathbf{f} = [f(x_1), \dots, f(x_\ell)]^T$ and Θ is the Vandermonde matrix associated with the nodes $\{x_\ell\}_{\ell=1}^m$. The traditional Lawson's iteration [12, 26], which consists of solving a sequence of weighted least-squares (LS) problems, is shown to be an effective method for solving the dual problem (2.16) (see [45]); it is an iteratively reweighted least-squares (IRLS) iteration [14, 31], and the involved weights are just the dual variables $\mathbf{w}$ which are updated according to the errors of the approximations from these weighted LS problems. Interestingly, as a generalization of [45, Theorem 2.1], for matrix-valued polynomial minimax approximations with $s \geq 1, t \geq 1$ and $d = 0$, Theorem 4.2 ensures that strong duality (2.17) is true.

- (2) For the rational minimax approximation with $s = t = 1$ and $d \geq 1$, [50, Section 4] connects strong duality (2.17) with Ruttan's sufficient condition [34, Theorem 2.1] (see also [37, Theorem 3]). In this case, though strong duality cannot be guaranteed for any situation of (1.2), it is true frequently in practice [50]. Theorem 4.1 establishes the counterpart of Ruttan's sufficient condition for matrix-valued rational minimax approximations.

3 Computation details for the dual function

3.1 Basic properties of the dual function $d(\mathbf{w})$

To compute the dual function value $d(\mathbf{w})$ at $\mathbf{w} \in \mathcal{S}$, we need to solve a constrained minimization (2.14). Fortunately, the dual formulation of the matrix-valued rational minimax approximation (2.14) takes a similar form as the one for scalar version (i.e., $s = t = 1$) given in [50]. Hence, the treatment in [50, Proposition 3.1] can be employed to transform it into a generalized eigenvalue problem, which is further related to finding the smallest singular pair of a tall matrix.

Proposition 3.1. *For $\mathbf{w} \in \mathcal{S}$, with the notation in (2.8), (2.9), (2.10) and (2.11), we have*

- (i) $\mathbf{c}(\mathbf{w}) = \begin{bmatrix} \mathbf{a}(\mathbf{w}) \\ \mathbf{b}(\mathbf{w}) \end{bmatrix} \in \mathbb{C}^{n+d+1}$ is a solution of (2.14) if and only if it is an eigenvector of the Hermitian positive semi-definite generalized eigenvalue problem $(A_{\mathbf{w}}, B_{\mathbf{w}})$ and $d(\mathbf{w})$ is the smallest eigenvalue satisfying

$$A_{\mathbf{w}}\mathbf{c}(\mathbf{w}) = d(\mathbf{w})B_{\mathbf{w}}\mathbf{c}(\mathbf{w}) \quad \text{and} \quad (\mathbf{c}(\mathbf{w}))^H B_{\mathbf{w}} \mathbf{c}(\mathbf{w}) = 1, \quad (3.1)$$

where

$$A_{\mathbf{w}} := [-\Theta, \mathbf{F}\Phi]^H \mathbf{W}_{\otimes} [-\Theta, \mathbf{F}\Phi] = \begin{bmatrix} \Theta^H \mathbf{W}_{\otimes} \Theta & -\Theta^H \mathbf{W}_{\otimes} \mathbf{F}\Phi \\ -\Phi^H \mathbf{F}^H \mathbf{W}_{\otimes} \Theta & \Phi^H \mathbf{F}^H \mathbf{W}_{\otimes} \mathbf{F}\Phi \end{bmatrix}, \quad (3.2)$$

$$B_{\mathbf{w}} := [0_{m \times n}, \Phi]^H W [0_{m \times n}, \Phi] = \begin{bmatrix} 0_{n \times n} & 0_{n \times (d+1)} \\ 0_{(d+1) \times n} & \Phi^H W \Phi \end{bmatrix} \in \mathbb{C}^{(n+d+1) \times (n+d+1)}, \quad (3.3)$$

- (ii) the Hermitian matrix $H_{\mathbf{w}} := A_{\mathbf{w}} - d(\mathbf{w})B_{\mathbf{w}} \succeq 0$, i.e., $H_{\mathbf{w}}$ is positive semi-definite;
(iii) let $\sqrt{W}\Phi = Q_q R_q \in \mathbb{C}^{m \times (d+1)}$ and $\sqrt{W_{\otimes}}\Theta = Q_p R_p \in \mathbb{C}^{gm \times n}$ be the thin QR factorizations where $Q_q \in \mathbb{C}^{m \times \tilde{n}_2}$, $Q_p \in \mathbb{C}^{gm \times \tilde{n}_1}$, $R_q \in \mathbb{C}^{\tilde{n}_2 \times (d+1)}$, $R_p \in \mathbb{C}^{\tilde{n}_1 \times n}$ with $\tilde{n}_1 = \text{rank}(\sqrt{W_{\otimes}}\Theta)$ and $\tilde{n}_2 = \text{rank}(\sqrt{W}\Phi)$. Then $d(\mathbf{w})$ is the smallest eigenvalue of the following Hermitian positive semi-definite matrix

$$S(\mathbf{w}) := S_F - S_{qp} S_{qp}^H \in \mathbb{C}^{\tilde{n}_2 \times \tilde{n}_2}, \quad (3.4)$$

where

$$S_F = Q_q^H \left(\sum_{j=1}^t \sum_{i=1}^s |F_{ij}|^2 \right) Q_q \in \mathbb{C}^{\tilde{n}_2 \times \tilde{n}_2}, \quad S_{qp} = Q_q^H \mathbf{F}^H Q_p \in \mathbb{C}^{\tilde{n}_2 \times \tilde{n}_1},$$

with $|F_{ij}| = \text{diag}(|f_{ij}(x_1)|, \dots, |f_{ij}(x_m)|) \in \mathbb{C}^{m \times m}$. Moreover, $\sqrt{d(\mathbf{w})}$ is the smallest singular value of $(I - Q_p Q_p^H) \mathbf{F} Q_q \in \mathbb{C}^{gm \times \tilde{n}_2}$ with the associated singular vector $\hat{\mathbf{b}} = R_q \mathbf{b}(\mathbf{w})$. Also, $R_p \mathbf{a}(\mathbf{w}) = S_{qp}^H \hat{\mathbf{b}}$.

Proof. With the notation in (2.8), (2.9), (2.10) and (2.11), the proof is the same as that for [50, Proposition 3.1], and the details are omitted. $\square$

For $\mathbf{w} \geq 0$ and $W = \text{diag}(\mathbf{w})$, we introduce the $\mathbf{w}$ -inner product (positive semidefinite): $\langle \mathbf{y}, \mathbf{z} \rangle_W = \mathbf{y}^H W \mathbf{z}$ and $\|\mathbf{y}\|_W = \sqrt{\mathbf{y}^H W \mathbf{y}}$, which, for simplicity, will also be written as $\langle \mathbf{y}, \mathbf{z} \rangle_{\mathbf{w}}$ and $\|\mathbf{y}\|_{\mathbf{w}}$, respectively. Using the $\mathbf{w}$ -inner product, we can rewrite the equations associated with the first n rows and the last $d+1$ rows of $A_{\mathbf{w}} \mathbf{c}(\mathbf{w}) = d(\mathbf{w}) B_{\mathbf{w}} \mathbf{c}(\mathbf{w})$ in (3.1) as the following optimality conditions:

Corollary 3.1. *Let $\mathbf{p} = \Theta \mathbf{a}(\mathbf{w}) = [\mathbf{p}_{11}^T, \dots, \mathbf{p}_{st}^T]^T \in \mathbb{C}^{mg}$ and $\mathbf{q} = \Phi \mathbf{b}(\mathbf{w}) \in \mathbb{C}^m$ be from the solution of (2.12) at $\mathbf{w} \in \mathcal{S}$, where $\mathbf{p}_{ij} = \Psi_{ij} \mathbf{a}_{ij} \in \mathbb{C}^m$ partitioned as in (2.10). Then for any $1 \leq i \leq s$ and $1 \leq j \leq t$, it holds that*

$$F_{ij} \mathbf{q} - \mathbf{p}_{ij} \perp_{\mathbf{w}} \text{span}(\Psi_{ij}), \quad \mathbf{F}^H (\mathbf{F} \mathbf{q} - \mathbf{p}) - d(\mathbf{w}) \mathbf{q} \perp_{\mathbf{w}} \text{span}(\Phi), \quad (3.5)$$

or equivalently, $\mathbf{F} \mathbf{q} - \mathbf{p} \perp_{W_{\otimes}} \text{span}(\Theta)$, $\mathbf{F}^H (\mathbf{F} \mathbf{q} - \mathbf{p}) - d(\mathbf{w}) \mathbf{q} \perp_W \text{span}(\Phi)$.

Furthermore, as a generalization of [48, Proposition 5.1], we have the explicit form of the gradient of $d(\mathbf{w})$.

Proposition 3.2. *For $\mathbf{w} > 0$, let $d(\mathbf{w})$ be the smallest eigenvalue of the Hermitian positive semi-definite generalized eigenvalue problem (3.4), and $\mathbf{c}(\mathbf{w}) = \begin{bmatrix} \mathbf{a}(\mathbf{w}) \\ \mathbf{b}(\mathbf{w}) \end{bmatrix} \in \mathbb{C}^{n+d+1}$ be the associated eigenvector. Denote $\mathbf{p} = [\mathbf{p}_{11}^T, \dots, \mathbf{p}_{st}^T]^T = \Theta \mathbf{a} \in \mathbb{C}^{mg}$ and $\mathbf{q} = \Phi \mathbf{b} \in \mathbb{C}^m$, where $\mathbf{p}_{ij} = \Psi_{ij} \mathbf{a}_{ij} \in \mathbb{C}^m$ partitioned as in (2.10). If $d(\mathbf{w})$ is a simple eigenvalue, then $d(\mathbf{w})$ is differentiable at $\mathbf{w}$ and its gradient is*

$$\begin{aligned} \nabla d(\mathbf{w}) &= \begin{bmatrix} \sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_1)q(x_1) - p_{ij}(x_1)|^2 - d(\mathbf{w})|q(x_1)|^2 \\ \sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_2)q(x_2) - p_{ij}(x_2)|^2 - d(\mathbf{w})|q(x_2)|^2 \\ \vdots \\ \sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_m)q(x_m) - p_{ij}(x_m)|^2 - d(\mathbf{w})|q(x_m)|^2 \end{bmatrix} \\ &= \sum_{i=1}^s \sum_{j=1}^t |F_{ij} \mathbf{q} - \mathbf{p}_{ij}|^2 - d(\mathbf{w}) |\mathbf{q}|^2 \in \mathbb{R}^m. \end{aligned} \quad (3.6)$$

Proof. See Appendix D. $\square$

3.2 Implementation based on Vandermonde with Arnoldi process (V+A)

Proposition 3.1, particularly item (iii), offers a way to compute the dual function $d(\mathbf{w})$. Represented in the monomial basis for p_{ij} ($1 \leq i \leq s$, $1 \leq j \leq t$) and q , the Vandermonde with Arnoldi process (V+A) presented in [7] is effective to handle the ill-conditionings in the involved Vandermonde basis matrices. V+A has been successfully employed in [25, 49, 50] to resolve the numerical stability in certain approximation problems.

In our case, we note that the Vandermonde matrix $\Phi = [\mathbf{1}_m, X\mathbf{1}_m, \dots, X^d\mathbf{1}_m]$ (i.e., $\phi_i(x) = x^{i-1}$, $i \geq 1$) with $X = \text{diag}(\mathbf{x}) \in \mathbb{C}^{m \times m}$ gives

$$\sqrt{W}\Phi = [\sqrt{\mathbf{w}}, X\sqrt{\mathbf{w}}, \dots, X^d\sqrt{\mathbf{w}}],$$

and the orthonormal basis Q_q of $\text{span}(\sqrt{W}\Phi)$ can be obtained via the Arnoldi process [7, 25, 49]. In particular, assuming $\text{rank}(\sqrt{W}\Phi) = d+1$ with $Q_q\mathbf{e}_1 = \frac{\sqrt{\mathbf{w}}}{\|\sqrt{\mathbf{w}}\|_2}$, the Arnoldi process produces (see [49, Theorem 2.1] and also [25])

$$XQ_q = Q_qH_q + \gamma_{d+1}\mathbf{q}_{d+2}\mathbf{e}_{d+1}^T, \quad \gamma_{d+1} \in \mathbb{C}, \quad (3.7)$$

where $H_q \in \mathbb{C}^{(d+1) \times (d+1)}$ is an upper Hessenberg matrix and $\mathbf{q}_{d+2} \perp Q_q$; also, the triangular matrix $R_q = \|\sqrt{\mathbf{w}}\|_2[\mathbf{e}_1, H_q\mathbf{e}_1, \dots, H_q^d\mathbf{e}_1]$ and $\sqrt{W}\Phi = Q_qR_q$ (see [49, Theorem 2.1]). Indeed, V+A produces a new basis (V+A basis) $\{\vartheta_j(x)\}_{j=0}^d$ for $\mathbb{P}_d$ satisfying the recurrence

$$x[\vartheta_0(x), \vartheta_1(x), \dots, \vartheta_d(x)] = [\vartheta_0(x), \vartheta_1(x), \dots, \vartheta_d(x), \vartheta_{d+1}(x)]H_q. \quad (3.8)$$

By using the computed H_d and γ_{d+1} in (3.7), the Vandermonde matrix $\Phi(y_1, \dots, y_{\widetilde{m}}; d)$ at new nodes $\{y_\ell\}_{\ell=1}^{\widetilde{m}}$ takes $\Phi(y_1, \dots, y_{\widetilde{m}}; d) = L_qR_q$ (see [49, Theorem 2.1]), where $L_q \in \mathbb{C}^{\widetilde{m} \times (d+1)}$ satisfies

$$YL_q = L_qH_q + \gamma_{d+1}\mathbf{l}_{d+2}\mathbf{e}_{d+1}^T, \quad L_q\mathbf{e}_1 = \mathbf{1}_{\widetilde{m}}, \quad Y = \text{diag}(y_1, \dots, y_{\widetilde{m}}). \quad (3.9)$$

Related with the numerator polynomial p_{ij} in the rational function $r_{ij} = p_{ij}/q$ and its Vandermonde matrix Ψ_{ij} given in (2.1), the same applies to $\sqrt{W}\Psi_{ij}$ to yield the orthonormal basis P_{ij} for $\text{span}(\sqrt{W}\Psi_{ij})$. In this procedure, letting $\nu = \max_{ij} n_{ij}$, we only need to compute the orthonormal basis P_ν for $\text{span}([\sqrt{\mathbf{w}}, X\sqrt{\mathbf{w}}, \dots, X^\nu\sqrt{\mathbf{w}}])$ via Arnoldi process [7, 25, 49], and set $P_{ij} = P_\nu(:, 1:n_{ij})$. Consequently, the matrices Q_p and R_p in the QR factorization of $\sqrt{W_\otimes}\Theta$ are

$$Q_p = \text{diag}(P_{11}, \dots, P_{s1}, P_{21}, \dots, P_{st}), \quad R_p = \text{diag}(R_{11}, \dots, R_{s1}, R_{21}, \dots, R_{st}).$$

With this and Proposition 3.1, $\sqrt{d(\mathbf{w})}$ is the smallest singular value of

$$(I - Q_p Q_p^H) \mathbf{F} Q_q = \begin{bmatrix} (I_{n_{11}} - P_{11} P_{11}^H) F_{11} Q_q \\ \vdots \\ (I_{n_{s1}} - P_{s1} P_{s1}^H) F_{s1} Q_q \\ (I_{n_{12}} - P_{12} P_{12}^H) F_{12} Q_q \\ \vdots \\ (I_{n_{st}} - P_{st} P_{st}^H) F_{st} Q_q \end{bmatrix} \in \mathbb{C}^{gm \times (d+1)} \quad (3.10)$$

and the associated singular vector $\hat{\mathbf{b}}$ satisfying $\hat{\mathbf{b}} = R_q \mathbf{b}(\mathbf{w})$. Indeed, this $\hat{\mathbf{b}}$ is the coefficient vector of the denominator q w.r.t. the V+A basis $\{\vartheta_j(x)\}_{j=0}^d$. Let

$$\hat{\mathbf{a}} := R_p \mathbf{a}(\mathbf{w}) = [(R_{11} \mathbf{a}_{11}(\mathbf{w}))^T, \dots, (R_{s1} \mathbf{a}_{s1}(\mathbf{w}))^T, (R_{12} \mathbf{a}_{12}(\mathbf{w}))^T, \dots, (R_{st} \mathbf{a}_{st}(\mathbf{w}))^T]^T.$$

By Proposition 3.1, $\hat{\mathbf{a}}$ can be computed according to $\hat{\mathbf{a}} = S_{qp}^H \hat{\mathbf{b}}$; denoting

$$\hat{\mathbf{a}} = [\hat{\mathbf{a}}_{11}^T, \dots, \hat{\mathbf{a}}_{s1}^T, \hat{\mathbf{a}}_{12}^T, \dots, \hat{\mathbf{a}}_{st}^T]^T, \quad \hat{\mathbf{a}}_{ij} \in \mathbb{C}^{n_{ij}+1},$$

we then have the correspondence $\hat{\mathbf{a}}_{ij} = R_{ij} \mathbf{a}_{ij}(\mathbf{w})$.

It is worthy mentioning that computing the vectors $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$ does not involve explicitly the upper triangular matrices R_p and R_q . Moreover, using $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$, we can compute vectors $\mathbf{p}_{ij}$ and $\mathbf{q}$ given in (2.2) and (2.3), respectively, as

$$\mathbf{p}_{ij} = \Psi_{ij} \mathbf{a}_{ij}(\mathbf{w}) = W^{-\frac{1}{2}} P_{ij} (R_{ij} \mathbf{a}_{ij}(\mathbf{w})) = W^{-\frac{1}{2}} P_{ij} \hat{\mathbf{a}}_{ij}, \quad (3.11a)$$

$$\mathbf{q} = \Phi \mathbf{b}(\mathbf{w}) = W^{-\frac{1}{2}} Q_q R_q \mathbf{b}(\mathbf{w}) = W^{-\frac{1}{2}} Q_q \hat{\mathbf{b}}, \quad (3.11b)$$

and hence $\mathbf{r}_{ij} = r_{ij}(\mathbf{x}) = [r_{ij}(x_1), \dots, r_{ij}(x_m)]^T = \mathbf{p}_{ij} / \mathbf{q} \in \mathbb{C}^m$.

For the evaluation values of $r_{ij}(x) = p_{ij}(x)/q(x)$ at new nodes $\{y_\ell\}_{\ell=1}^{\widetilde{m}}$, V+A relies on (3.8), (3.9) and $\Phi(y_1, \dots, y_{\widetilde{m}}; d) = L_q R_q$ to get

$$q(\widetilde{\mathbf{y}}) = [q(y_1), \dots, q(y_{\widetilde{m}})]^T = \Phi(y_1, \dots, y_{\widetilde{m}}; d) R_q^{-1} (R_q \mathbf{b}(\mathbf{w})) = L_q \hat{\mathbf{b}}; \quad (3.12)$$

the same applies to the numerator polynomial $p_{ij}(x)$ to get

$$p_{ij}(\widetilde{\mathbf{y}}) = [p_{ij}(y_1), \dots, p_{ij}(y_{\widetilde{m}})]^T = \Psi_{ij}(y_1, \dots, y_{\widetilde{m}}; n_{ij}) R_{ij}^{-1} (R_{ij} \mathbf{a}_{ij}(\mathbf{w})) = L_{ij} \hat{\mathbf{a}}_{ij}. \quad (3.13)$$

Thus it leads to the values of $\{r_{ij}(y_\ell)\}_{\ell=1}^{\widetilde{m}}$ and the vectors $r_{ij}(\mathbf{y}) = p_{ij}(\mathbf{y}) / q(\mathbf{y}) \in \mathbb{C}^{\widetilde{m}}$. This technique has previously been used in, e.g., [25, 45, 50].

4 Strong duality

In Section 2.2, we have listed some special cases for which strong duality (2.17) holds, and thereby, the original minimax approximations can be recovered from solving the

dual problem (2.16) in these cases. For the scalar version of rational minimax approximation (i.e., $s = t = 1$ and $d \geq 1$), [50] has shown that, strong duality (2.17) is equivalent to the classical Ruttan's sufficient condition [50, Equ. (4.4)] (see also [37, Theorem 4]). In this section, we shall continue this issue by providing an easily checkable condition for strong duality (2.17). This condition uses the information of the solution of the dual problem (2.16), and can be viewed as the generalized Ruttan's sufficient condition for the matrix-valued rational minimax approximation (1.2).

Theorem 4.1. *Let $\mathbf{w}^* \in \mathcal{S}$ be the maximizer of the dual problem (2.16), and $(\mathbf{a}^*, \mathbf{b}^*)$ be the associated solution of (2.14) that achieves the minimum $d(\mathbf{w}^*)$, where $\mathbf{a}^*$ contains the coefficient vectors $\mathbf{a}_{ij}^* \in \mathbb{C}^{n_{ij}+1}$ partitioned as in (2.10). Denote*

$$R^*(x) = \begin{bmatrix} r_{11}^*(x) & \cdots & r_{1t}^*(x) \\ \vdots & \ddots & \vdots \\ r_{s1}^*(x) & \cdots & r_{st}^*(x) \end{bmatrix}, \quad r_{ij}^*(x) = \frac{p_{ij}^*(x)}{q^*(x)}$$

with $p_{ij}^*(x) = [\psi_0(x), \dots, \psi_{ij}(x)]\mathbf{a}_{ij}^*$ and $q(x) = [\phi_0(x), \dots, \phi_d(x)]\mathbf{b}^*$. Suppose $q^*(x_\ell) \neq 0$ for all $x_\ell \in \mathcal{X}$. Then if

$$d(\mathbf{w}^*) = \max_{x_\ell \in \mathcal{X}} \|F(x_\ell) - R^*(x_\ell)\|_F^2 := e(R^*), \quad (4.1)$$

then strong duality (2.17) holds and R^* is the global solution to (1.2).

Proof. See Appendix E. □

We remark that (4.1) is a sufficient condition to guarantee the solvability of (1.2) and strong duality (2.17). The following theorem shows that for the matrix-valued polynomial minimax approximation with $s \geq 1, t \geq 1$ and $d = 0$ in (1.2), strong duality (2.17) holds.

Theorem 4.2. *For the matrix-valued polynomial minimax approximation with $s \geq 1, t \geq 1$ and $d = 0$ in (1.3), strong duality (2.17) is true and the matrix-valued minimax polynomial P^* of (1.2) can be solved from the dual problem (2.16).*

Proof. For the polynomial case, we first note that (1.2) is equivalent to

$$\begin{aligned} & \inf_{\eta \in \mathbb{R}, p_{ij} \in \mathbb{P}_{n_{ij}}} \eta \\ & \text{s.t.}, \|F(x_\ell) - P(x_\ell)\|_F^2 \leq \eta, \quad 1 \leq \ell \leq m, \end{aligned} \quad (4.2)$$

where $P : \mathbb{C} \rightarrow \mathbb{C}^{s \times t}$ is the matrix-valued polynomial with $p_{ij} \in \mathbb{P}_{n_{ij}}$ as its (i, j) th component. Using notation in (2.9), we know that each constraint in (4.2) can be written as

$$\|\mathbf{F}\mathbf{e}_\ell - \Pi_\ell \Theta \mathbf{a}\|_2^2 - \eta \leq 0,$$

which is convex with respect to $\mathbf{a}$ and η , where

$$\Pi_\ell = \text{diag}(\underbrace{\mathbf{e}_\ell \mathbf{e}_\ell^T, \dots, \mathbf{e}_\ell \mathbf{e}_\ell^T}_g) \in \mathbb{R}^{mg \times mg}, \quad \mathbf{e}_\ell \in \mathbb{R}^m.$$

Therefore, (4.2) is a convex minimization and the infimum is achievable. Note that (2.16) is the dual problem of (4.2). Also, the Slater condition (see e.g., [6, Section 5.2.3]) holds for (4.2) since there is $(\eta, \mathbf{a})$ so that $\|\mathbf{F}\mathbf{e}_\ell - \Pi_\ell \Theta \mathbf{a}\|_2^2 - \eta < 0$ for any $\ell = 1, 2, \dots, m$. Hence, strong duality (2.17) is ensured in this case by [6, Section 5.2.3]. $\square$

In the framework of the dual problem (2.16), we point out that the condition (4.1) for strong duality (2.17) is numerically checkable. In particular, when an approximation $\mathbf{w} \in \mathcal{S}$ of the global maximizer of the dual (2.16) together with the associated $R(x)$ is available, we then can compute the relative error of strong duality

$$\epsilon(\mathbf{w}) := \left| \frac{e(R) - d(\mathbf{w})}{e(R)} \right|, \quad \text{where } e(R) = \max_{x_\ell \in \mathcal{X}} \|F(x_\ell) - R(x_\ell)\|_{\mathbb{F}}^2, \quad (4.3)$$

as a measure of the accuracy of the rational approximation $R(x)$.

5 Extreme points and complementary slackness

A noticeable property, known as the equioscillation property [40, Theorem 24.1], of the real polynomial minimax approximation describes the alternation of the maximum error occurring in the so-called *extreme points* (aka *reference points*). In our case, particularly, for the solution $R^* : \mathbb{C} \rightarrow \mathbb{C}^{s \times t}$ of (1.2), extreme points refer to nodes that achieve $\|F(x_\ell) - R^*(x_\ell)\|_{\mathbb{F}}^2 = e(R^*) := \max_{x_j \in \mathcal{X}} \|F(x_j) - R^*(x_j)\|_{\mathbb{F}}^2$. Those points are denoted by

$$\mathcal{X}_e(R^*) := \{x_\ell \in \mathcal{X} : \|F(x_\ell) - R^*(x_\ell)\|_{\mathbb{F}}^2 = e(R^*)\}. \quad (5.1)$$

In the following theorem, under strong duality (2.17), we prove a complementary slackness and also show that non-extreme points in $\mathcal{X}$ can be filtered out to reduce the computational costs and accelerate the convergence for solving the dual problem (2.16). This is a generalization of [50, Theorem 2.4] to the matrix-valued functions. The same strategy has been used in Lawson's iteration for the linear [8, 26, 33, 45] and rational [50] minimax approximations.

Theorem 5.1. *Suppose $R^*(x) = [r_{ij}^*(x)] \in \mathcal{R}_{(s,t)}$ is a solution to (1.2) with $q^*(x_j) \neq 0$ ($\forall x_j \in \mathcal{X}$). If strong duality (2.17) holds, then*

(i) *for any solution $\mathbf{w}^*$ of the dual problem (2.16) with $d(\mathbf{w}^*)$ given in (2.14), we have*

$$(\text{complementary slackness}) \quad w_\ell^*(e(R^*) - \|F(x_\ell) - R^*(x_\ell)\|_{\mathbb{F}}^2) = 0, \quad \forall 1 \leq \ell \leq m; \quad (5.2)$$

(ii) *(number of extreme points) there are at least $\min_{1 \leq i \leq s, 1 \leq j \leq t} (n_{ij} + 2)$ extreme points associated with $R^*(x)$, i.e.,*

$$|\mathcal{X}_e(R^*)| \geq \min_{1 \leq i \leq s, 1 \leq j \leq t} (n_{ij} + 2); \quad (5.3)$$

- (iii) (filtering out non-extreme points) for any subset $\check{\mathcal{X}} \subseteq \mathcal{X}$ satisfying $\mathcal{X}_e(R^*) \subseteq \check{\mathcal{X}}$, $(\eta_\infty, \{p_{ij}^*\}, q^*)$ solves

$$\begin{aligned} & \inf_{\eta \in \mathbb{R}, p_{ij} \in \mathbb{P}_{n_{ij}}, 0 \neq q \in \mathbb{P}_d} \eta \\ \text{s.t.}, & \sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_\ell)q(x_\ell) - p_{ij}(x_\ell)|^2 \leq \eta |q(x_\ell)|^2, \quad x_\ell \in \check{\mathcal{X}} \end{aligned} \quad (5.4)$$

for which strong duality holds too; that is,

$$\eta_\infty = \max_{\check{\mathbf{w}} \in \check{\mathcal{S}}} \check{d}(\check{\mathbf{w}}), \quad (5.5)$$

where $\check{\mathcal{S}} = \{\check{\mathbf{w}} = [\check{w}_1, \dots, \check{w}_s]^T \in \mathbb{R}^{\check{s}} : \check{\mathbf{w}} \geq 0 \text{ and } \check{\mathbf{w}}^T \mathbf{e} = 1\}$, $\check{s} = |\check{\mathcal{X}}|$, and

$$\check{d}(\check{\mathbf{w}}) = \min_{\substack{p_{ij} \in \mathbb{P}_{n_{ij}}, q \in \mathbb{P}_d \\ \sum_{x_\ell \in \check{\mathcal{X}}} \check{w}_\ell |q(x_\ell)|^2 = 1}} \sum_{x_\ell \in \check{\mathcal{X}}} \check{w}_\ell \sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_\ell)q(x_\ell) - p_{ij}(x_\ell)|^2. \quad (5.6)$$

Moreover, for any solution $\check{\mathbf{w}}^*$ of the dual (5.5), $(\{p_{ij}^*\}, q^*)$ achieves the minimum $\check{d}(\check{\mathbf{w}}^*) = \eta_\infty$ of (5.6).

Proof. See Appendix F. □

As a consequence of Theorem 4.2 and item (ii) of Theorem 5.1, we have

Corollary 5.1. For the matrix-valued polynomial minimax approximation with $s \geq 1, t \geq 1$ and $d = 0$ in (1.3), $|\mathcal{X}_e(P^*)| \geq \min_{1 \leq i \leq s, 1 \leq j \leq t} (n_{ij} + 2)$ for the solution P^* of (1.2).

6 m-d-Lawson: a method for rational minimax approximation of matrix-valued functions

Lawson's iteration [26] is a traditional method for solving the discrete linear minimax approximations of complex-valued functions. Besides some previous convergence of Lawson's iteration for the polynomial minimax approximation (see e.g., [3, 8, 12, 33]), recent works in [45, 48, 50] have revealed that Lawson's iteration is indeed a method for solving the associated dual problems. Closely relying upon this connection, [48, 50] proposed a dual Lawson's iteration, namely, **d-Lawson**, for the discrete rational minimax approximation, and established basic convergence analysis. Along this way, we propose a dual Lawson's iteration (Algorithm 1) for the matrix-valued rational minimax approximation (1.2). In our paper, this algorithm is termed **m-d-Lawson**, indicating its dual-based **Lawson** iteration for **matrix-valued** functions.

In each iteration, **m-d-Lawson** consists of a calling of a thin SVD in Step 3 (see (3.10)) and a basic updating of the weight vector $\mathbf{w}$ in Step 5. The parameter $\beta > 0$ involving in Step 5 is called the *Lawson exponent* [8, 26] and is a factor for convergence.

Algorithm 1 A matrix-valued dual Lawson's iteration (**m-d-Lawson**) for (1.2)

Input: Given $\{(x_\ell, F(x_\ell))\}_{\ell=1}^m$ with $x_\ell \in \Omega$, a tolerance for strong duality $\epsilon_r > 0$, the maximum number $k_{\maxit}$ of iterations

Output: An approximant $R^* \in \mathcal{R}_{(s,t)}$

- 1: (Initialization) Let $k = 0$; choose $\mathbf{0} < \mathbf{w}^{(0)} \in \mathbb{S}$ and a tolerance ϵ_w for the weights;
- 2: (Filtering: optional) Remove nodes x_ℓ with $w_\ell^{(k)} < \epsilon_w$;
- 3: Compute $d(\mathbf{w}^{(k)})$ and the associated $r_{ij}^{(k)}(\mathbf{x}) = \mathbf{p}_{ij}^{(k)} \cdot \mathbf{q}^{(k)}$ according to (3.11);
- 4: (Stop rule) Stop and return $R^* = R^{(k)}$, either if $k \geq k_{\maxit}$ or

$$\epsilon(\mathbf{w}^{(k)}) := \left| \frac{e(R^{(k)}) - d(\mathbf{w}^{(k)})}{e(R^{(k)})} \right| < \epsilon_r, \quad \text{where } e(R^{(k)}) = \max_{x_\ell \in \mathcal{X}} \|F(x_\ell) - R^{(k)}(x_\ell)\|_{\mathbb{F}}^2; \quad (6.1)$$

- 5: (Updating weights) Update $\mathbf{w}^{(k+1)}$ according to $(\beta > 0)$

$$w_\ell^{(k+1)} = \frac{w_\ell^{(k)} \|F(x_\ell) - R^{(k)}(x_\ell)\|_{\mathbb{F}}^\beta}{\sum_i w_i^{(k)} \|F(x_i) - R^{(k)}(x_i)\|_{\mathbb{F}}^\beta}, \quad \forall \ell,$$

and go to Step 2 with $k = k + 1$.

In our numerical testing, we choose $\beta = 1$, and set $\epsilon_w = 0, \epsilon_r = 10^{-3}$ in Step 2 and Step 4, respectively.

Regarding the convergence of **m-d-Lawson**, we prove in Section 7 that for polynomial minimax approximations, the choice $\beta = 1$ is near-optimal, and the sequence $\{d(\mathbf{w}^{(k)})\}$ monotonically increases for any $\beta \in (0, 2)$; for rational minimax approximations, any sufficiently small $\beta > 0$ generically ensures the monotonic convergence. These results generalize the scalar-case ($s = t = 1$) convergence properties established in [48].

7 Convergence analysis of m-d-Lawson

Throughout this section, we set $\epsilon_w = 0$ in Step 2 of Algorithm 1 and also assume the following two fundamental conditions:

- (A1) $q_\ell^{(k)} := \mathbf{e}_\ell^T \mathbf{q}^{(k)} = q^{(k)}(x_\ell) \neq 0$ for any $1 \leq \ell \leq m$ and any $k \geq 0$;
- (A2) the cardinality $|\mathcal{J}_{\mathbf{w}^{(k)}}| \geq \max\{d + 1, \max_{1 \leq i \leq s, 1 \leq j \leq t} \{n_{ij} + 1\}\}$, where $\mathcal{J}_{\mathbf{w}^{(k)}} := \{\ell : w_\ell^{(k)} \neq 0, 1 \leq \ell \leq m\}$.

Proposition 7.1. *If $d(\mathbf{w}^{(k)}) > 0$, then*

$$\mathcal{J}_{\mathbf{w}^{(k+1)}} = \{\ell : w_\ell^{(k)} \tau_\ell^{(k)} \neq 0, 1 \leq \ell \leq m\}, \quad \text{where } \tau_\ell^{(k)} := \|F(x_\ell) - R^{(k)}(x_\ell)\|_{\mathbb{F}}. \quad (7.1)$$

Proof. By (A1), $d(\mathbf{w}^{(k)}) > 0$ implies $\sum_{\ell=1}^m w_\ell^{(k)} \|F(x_j) - R^{(k)}(x_\ell)\|_F^\beta > 0$. According to Lawson's updating in Step 5 of Algorithm 1, (7.1) holds as $w_\ell^{(k+1)} = 0 \iff w_\ell^{(k)} \tau_\ell^{(k)} = 0$. $\square$

7.1 A lower bound of the dual objective function value

From the perspective of dual problem (2.16) optimization, Lawson's iteration can be interpreted as a gradient ascent method. Indeed, suppose the current $\mathbf{0} < \mathbf{w} \in \mathcal{S}$, $q_j = \mathbf{q}^T \mathbf{e}_j \neq 0$, $\forall 1 \leq j \leq m$ and $d(\mathbf{w})$ is the simple eigenvalue of the matrix pencil $(A_{\mathbf{w}}, B_{\mathbf{w}})$; then

$$\mathbf{g}(\mathbf{w}) := \text{diag}\left(\frac{w_1}{|q_1|^2}, \dots, \frac{w_m}{|q_m|^2}\right) \nabla d(\mathbf{w}) \in \mathbb{R}^m$$

is an ascent direction for the dual function $d(\mathbf{w})$ because by Proposition 3.2

$$\mathbf{g}(\mathbf{w})^T \nabla d(\mathbf{w}) = \nabla d(\mathbf{w})^T \text{diag}\left(\frac{w_1}{|q_1|^2}, \dots, \frac{w_m}{|q_m|^2}\right) \nabla d(\mathbf{w}) > 0.$$

Relying on this direction and the step-size $s = \frac{1}{d(\mathbf{w})} > 0$, the update

$$\mathbf{w} + s\mathbf{g} = \frac{1}{d(\mathbf{w})} \begin{bmatrix} w_1 \|F(x_1) - R(x_1)\|_F^2 \\ \vdots \\ w_m \|F(x_m) - R(x_m)\|_F^2 \end{bmatrix}$$

can be scaled to have $\tilde{\mathbf{w}} = \frac{\mathbf{w} + s\mathbf{g}}{\mathbf{e}^T(\mathbf{w} + s\mathbf{g})} \in \mathcal{S}$. This says that an iteration of m-d-Lawson in Algorithm 1 with the Lawson exponent $\beta = 2$ is an ascent projection gradient iteration. While this connection provides valuable insight into m-d-Lawson, proving the precise monotonic convergence of the sequence $\{d(\mathbf{w}^{(k)})\}_k$ requires further investigation. This constitutes the main goal of this section.

Lemma 7.1. *Let $(\mathbf{a}^{(k)}, \mathbf{b}^{(k)})$ be the solution to (2.12) at $\mathbf{w}^{(k)} \in \mathcal{S}$ and $(\{p_{ij}^{(k)}\}, q^{(k)})$ be the associated polynomials. Assume (A1) and (A2). Let $\mathbf{p}^{(k)} = \Theta \mathbf{a}^{(k)}$, $\mathbf{q}^{(k)} = \Phi \mathbf{b}^{(k)}$, $\mathbf{1}_g = [1, \dots, 1]^T \in \mathbb{R}^g$ with $g = st$, and*

$$\boldsymbol{\tau}^{(k)}(\beta) := [(\tau_1^{(k)})^\beta, \dots, (\tau_m^{(k)})^\beta]^T \in \mathbb{R}^m,$$

where $\tau_\ell^{(k)} = \|F(x_\ell) - R^{(k)}(x_\ell)\|_F$ ($1 \leq \ell \leq m$) and $R^{(k)}(x) = [p_{ij}^{(k)}(x)/q^{(k)}(x)] : \mathbb{C} \rightarrow \mathbb{C}^{s \times t}$. If $d(\mathbf{w}^{(k)}) > 0$, then

$$\frac{(\mathbf{F}\mathbf{q}^{(k)} - \mathbf{p}^{(k)})}{\mathbf{1}_g \otimes \boldsymbol{\tau}^{(k)}(\beta)} \perp_{\mathbf{W}_{\otimes}^{(k+1)}} \text{span}(\Theta), \quad (7.2)$$

where $\mathbf{W}_{\otimes}^{(k+1)}$ is defined by (2.11) and we set $0_0^0 = 0$ for convenience.

Proof. Denote $\mathbf{u} = \frac{(\mathbf{F}\mathbf{q}^{(k)} - \mathbf{p}^{(k)})}{\mathbf{1}_g \otimes \boldsymbol{\tau}^{(k)}(\beta)}$ and

$$\gamma_\beta^{(k)} := \sum_{\ell=1}^m w_\ell^{(k)} \|F(x_\ell) - R^{(k)}(x_\ell)\|_F^\beta = \sum_{\ell=1}^m w_\ell^{(k)} (\tau_\ell^{(k)})^\beta. \quad (7.3)$$

To simplify notation, we drop the iteration superscript (e.g. $\mathbf{p} \equiv \mathbf{p}^{(k)}$ and $\boldsymbol{\tau} \equiv \boldsymbol{\tau}^{(k)}$) and use tilded notation for the next iterate (e.g. $\tilde{\mathbf{p}} \equiv \mathbf{p}^{(k+1)}$). This convention extends to all variables.

Since $d(\mathbf{w}) > 0$, $\gamma_\beta \neq 0$. For $\forall \mathbf{t} = [\mathbf{t}_{11}^T, \dots, \mathbf{t}_{st}^T]^T \in \text{span}(\boldsymbol{\Theta})$ with $\mathbf{t}_{ij} \in \mathbb{C}^m$ partitioned as in (2.10), by Corollary 3.1, we have

$$\begin{aligned} 0 &= \langle \mathbf{t}, \mathbf{F}\mathbf{q} - \mathbf{p} \rangle_{\mathbf{W}_\otimes} \\ &= \mathbf{t}^H \mathbf{W}_\otimes (\mathbf{F}\mathbf{q} - \mathbf{p}) = \sum_{i=1}^s \sum_{j=1}^t \mathbf{t}_{ij}^H W(F_{ij}\mathbf{q} - \mathbf{p}_{ij}) \\ &= \sum_{\ell=1}^m w_\ell \left(\sum_{i=1}^s \sum_{j=1}^t (\mathbf{e}_\ell^T \bar{\mathbf{t}}_{ij}) (f_{ij}(x_\ell)q(x_\ell) - p_{ij}(x_\ell)) \right). \end{aligned} \quad (7.4)$$

According to Lawson's updating rule in Algorithm 1, we have

$$\widetilde{\mathbf{W}_\otimes} = \frac{1}{\gamma_\beta} (I_g \otimes \text{diag}(\boldsymbol{\tau}(\beta))) \mathbf{W}_\otimes, \quad \tilde{w}_\ell = \frac{w_\ell \tau_\ell^\beta}{\gamma_\beta} \quad (1 \leq \ell \leq m).$$

By $\tau_\ell^2 = \|F(x_\ell) - R(x_\ell)\|_F^2 = \frac{1}{|q(x_\ell)|^2} \sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_\ell)q(x_\ell) - p_{ij}(x_\ell)|^2$ and $q(x_\ell) \neq 0$, a node x_ℓ satisfying $\tau_\ell = 0$ implies $f_{ij}(x_\ell)q(x_\ell) = p_{ij}(x_\ell)$ for all pairs (i, j) with $1 \leq i \leq s, 1 \leq j \leq t$. Also, $w_\ell = 0$ implies $\tilde{w}_\ell = 0$. Thus, based on (7.4), nodes x_ℓ with $w_\ell \tau_\ell = 0$, i.e., $\ell \notin \mathcal{J}_{\tilde{\mathbf{w}}}$ in (7.1), satisfy $\tilde{w}_\ell = 0$ and contribute nothing to $\langle \mathbf{t}, \mathbf{F}\mathbf{q} - \mathbf{p} \rangle_{\mathbf{W}_\otimes}$. Consequently, we obtain from (7.4) that

$$\begin{aligned} 0 &= \langle \mathbf{t}, \mathbf{F}\mathbf{q} - \mathbf{p} \rangle_{\mathbf{W}_\otimes} = \sum_{\ell \in \mathcal{J}_{\tilde{\mathbf{w}}}} w_\ell \left(\sum_{i=1}^s \sum_{j=1}^t (\mathbf{e}_\ell^T \bar{\mathbf{t}}_{ij}) (f_{ij}(x_\ell)q(x_\ell) - p_{ij}(x_\ell)) \right) \\ &= \sum_{\ell \in \mathcal{J}_{\tilde{\mathbf{w}}}} \frac{\tilde{w}_\ell \gamma_\beta}{\tau_\ell^\beta} \left(\sum_{i=1}^s \sum_{j=1}^t (\mathbf{e}_\ell^T \bar{\mathbf{t}}_{ij}) (f_{ij}(x_\ell)q(x_\ell) - p_{ij}(x_\ell)) \right) \\ &= \gamma_\beta \sum_{\ell \in \mathcal{J}_{\tilde{\mathbf{w}}}} \tilde{w}_\ell \left(\sum_{i=1}^s \sum_{j=1}^t (\mathbf{e}_\ell^T \bar{\mathbf{t}}_{ij}) \frac{f_{ij}(x_\ell)q(x_\ell) - p_{ij}(x_\ell)}{\tau_\ell^\beta} \right) \\ &= \gamma_\beta \sum_{\ell=1}^m \tilde{w}_\ell \left(\sum_{i=1}^s \sum_{j=1}^t (\mathbf{e}_\ell^T \bar{\mathbf{t}}_{ij}) \frac{f_{ij}(x_\ell)q(x_\ell) - p_{ij}(x_\ell)}{\tau_\ell^\beta} \right) \\ &= \gamma_\beta \mathbf{t}^H \widetilde{\mathbf{W}_\otimes} \mathbf{u} = \gamma_\beta \langle \mathbf{t}, \mathbf{u} \rangle_{\widetilde{\mathbf{W}_\otimes}}. \end{aligned} \quad (7.5)$$

In deriving (7.5), we retain summations over nodes x_ℓ where $\tilde{w}_\ell = 0$ (i.e., $\ell \notin \mathcal{J}_{\tilde{\mathbf{w}}}$). These nodes correspond to either $w_\ell = 0$ or $\tau_\ell = |f_{ij}(x_\ell)q(x_\ell) - p_{ij}(x_\ell)| = 0$ for all (i, j) . For the latter scenario, we adopt the convention $0_0^0 = 0$ for terms of the form $(f_{ij}(x_\ell)q(x_\ell) - p_{ij}(x_\ell))/\tau_\ell^\beta$. $\square$

We next provide a relation between $d(\mathbf{w}^{(k)})$ and $d(\mathbf{w}^{(k+1)})$.

Theorem 7.1. *Using the same notation and assumptions in Lemma 7.1, it holds that*

$$\sqrt{d(\mathbf{w}^{(k+1)})} \geq d(\mathbf{w}^{(k)}) \frac{|(\mathbf{q}^{(k+1)})^H \mathbf{W}^{(k)} \mathbf{q}^{(k)}|}{\gamma_\beta^{(k)} \zeta_\beta^{(k)}}, \quad (7.6)$$

where $\mathbf{W}^{(k)} = \text{diag}(\mathbf{w}^{(k)})$, $\gamma_\beta^{(k)}$ is given in (7.3) and

$$\zeta_\beta^{(k)} = \left\| \frac{(\mathbf{F}\mathbf{q}^{(k)} - \mathbf{p}^{(k)})}{\mathbf{1}_g \otimes \boldsymbol{\tau}^{(k)}(\beta)} \right\|_{\mathbf{W}_\otimes^{(k+1)}}, \quad \mathbf{W}_\otimes^{(k+1)} = I_g \otimes \mathbf{W}^{(k+1)}.$$

Proof. We consider only the case where $d(\mathbf{w}^{(k)}) > 0$, as (7.6) holds trivially otherwise. Omit the subscript k in each quantity related with the k th iteration as in the proof of Lemma 7.1.

Recall (7.1): $\mathcal{J}_{\tilde{\mathbf{w}}} = \{\ell : \tilde{w}_\ell \neq 0, 1 \leq \ell \leq m\} = \{\ell : w_\ell \tau_\ell \neq 0, 1 \leq \ell \leq m\}$. Since $d(\tilde{\mathbf{w}}) = \|\mathbf{F}\tilde{\mathbf{q}} - \tilde{\mathbf{p}}\|_{\tilde{\mathbf{W}}_\otimes}^2$, and $\mathbf{F}\tilde{\mathbf{q}} - \tilde{\mathbf{p}} \perp_{\tilde{\mathbf{W}}_\otimes} \tilde{\mathbf{p}}$ (note $\tilde{\mathbf{p}} \in \text{span}(\boldsymbol{\Theta})$) by Corollary 3.1, we get

$$0 \leq d(\tilde{\mathbf{w}}) = \langle \mathbf{F}\tilde{\mathbf{q}} - \tilde{\mathbf{p}}, \mathbf{F}\tilde{\mathbf{q}} \rangle_{\tilde{\mathbf{W}}_\otimes} = \sqrt{d(\tilde{\mathbf{w}})} \left\langle \frac{\mathbf{F}\tilde{\mathbf{q}} - \tilde{\mathbf{p}}}{\|\mathbf{F}\tilde{\mathbf{q}} - \tilde{\mathbf{p}}\|_{\tilde{\mathbf{W}}_\otimes}}, \mathbf{F}\tilde{\mathbf{q}} \right\rangle_{\tilde{\mathbf{W}}_\otimes},$$

which implies $\sqrt{d(\tilde{\mathbf{w}})} = \left\langle \frac{\mathbf{F}\tilde{\mathbf{q}} - \tilde{\mathbf{p}}}{\|\mathbf{F}\tilde{\mathbf{q}} - \tilde{\mathbf{p}}\|_{\tilde{\mathbf{W}}_\otimes}}, \mathbf{F}\tilde{\mathbf{q}} \right\rangle_{\tilde{\mathbf{W}}_\otimes}$.

Let

$$\mathbf{u} = \frac{(\mathbf{F}\mathbf{q} - \mathbf{p})}{\mathbf{1}_g \otimes \boldsymbol{\tau}(\beta)} = [\mathbf{u}_{11}^T, \dots, \mathbf{u}_{st}^T]^T \text{ with } \mathbf{u}_{ij} = \frac{(F_{ij}\mathbf{q} - \mathbf{p}_{ij})}{\boldsymbol{\tau}(\beta)} \in \mathbb{C}^m;$$

then (7.2) says $\mathbf{u} \perp_{\tilde{\mathbf{W}}_\otimes} \text{span}(\boldsymbol{\Theta})$. According to Corollary 3.1, it implies that

$$\tilde{\mathbf{p}} \in \underset{\mathbf{z} \in \text{span}(\boldsymbol{\Theta})}{\text{argmin}} \|\mathbf{F}\tilde{\mathbf{q}} - \mathbf{z}\|_{\tilde{\mathbf{W}}_\otimes}^2. \quad (7.7)$$

Note that $\tilde{\mathbf{W}}_\otimes = \text{diag}(\mathbf{1}_g \otimes \tilde{\mathbf{w}})$. By assumptions (A1) and (A2), the cardinality of the index set $\mathcal{J}_{\mathbf{1}_g \otimes \tilde{\mathbf{w}}}$ consisting of positive entries of $\mathbf{1}_g \otimes \tilde{\mathbf{w}}$ satisfies

$$|\mathcal{J}_{\mathbf{1}_g \otimes \tilde{\mathbf{w}}}| \geq \sum_{i=1}^s \sum_{j=1}^t (n_{ij} + 1) = n.$$

On the other hand, it is easy to see from assumptions **(A1)**, **(A2)** and (2.9) that

$$\text{rank}(\Theta(\mathcal{J}_{\mathbf{1}_g \otimes \tilde{\mathbf{w}}}, \cdot)) = \sum_{i=1}^s \sum_{j=1}^t \text{rank}(\Psi_{ij}(\mathcal{J}_{\tilde{\mathbf{w}}}, \cdot)) = \sum_{i=1}^s \sum_{j=1}^t (n_{ij} + 1) = n.$$

Thus applying [48, Lemma 3.1] to the least-square problem (7.7) and using $\mathbf{u} \perp_{\widetilde{\mathbf{W}}_\otimes} \text{span}(\Theta)$, we have

$$\sqrt{d(\tilde{\mathbf{w}})} = \left\langle \frac{\mathbf{F}\tilde{\mathbf{q}} - \tilde{\mathbf{p}}}{\|\mathbf{F}\tilde{\mathbf{q}} - \tilde{\mathbf{p}}\|_{\widetilde{\mathbf{W}}_\otimes}}, \mathbf{F}\tilde{\mathbf{q}} \right\rangle_{\widetilde{\mathbf{W}}_\otimes} \geq \left| \left\langle \mathbf{F}\tilde{\mathbf{q}}, \frac{\mathbf{u}}{\|\mathbf{u}\|_{\widetilde{\mathbf{W}}_\otimes}} \right\rangle_{\widetilde{\mathbf{W}}_\otimes} \right|. \quad (7.8)$$

Next, we show $\left| \langle \mathbf{F}\tilde{\mathbf{q}}, \mathbf{u} \rangle_{\widetilde{\mathbf{W}}_\otimes} \right| = \frac{d(\mathbf{w})}{\gamma_\beta} \left| \tilde{\mathbf{q}}^H \mathbf{W} \mathbf{q} \right|$, which together with (7.8) then leads to the desired result (7.6). Indeed,

$$\begin{aligned} \left| \langle \mathbf{F}\tilde{\mathbf{q}}, \mathbf{u} \rangle_{\widetilde{\mathbf{W}}_\otimes} \right| - \frac{d(\mathbf{w})}{\gamma_\beta} \left| \tilde{\mathbf{q}}^H \mathbf{W} \mathbf{q} \right| &= \left| (\mathbf{F}\tilde{\mathbf{q}})^H \widetilde{\mathbf{W}}_\otimes \frac{(\mathbf{F}\mathbf{q} - \mathbf{p})}{\mathbf{1}_g \otimes \tau(\beta)} \right| - \frac{d(\mathbf{w})}{\gamma_\beta} \left| \tilde{\mathbf{q}}^H \mathbf{W} \mathbf{q} \right| \\ &= \frac{1}{\gamma_\beta} \left(\left| \tilde{\mathbf{q}}^H \mathbf{W} \mathbf{F}^H (\mathbf{F}\mathbf{q} - \mathbf{p}) \right| - d(\mathbf{w}) \left| \tilde{\mathbf{q}}^H \mathbf{W} \mathbf{q} \right| \right). \end{aligned}$$

Since $\tilde{\mathbf{q}} \in \text{span}(\Phi)$, by the second optimality (3.5) in Corollary 3.1,

$$\mathbf{F}^H (\mathbf{F}\mathbf{q} - \mathbf{p}) - d(\mathbf{w})\mathbf{q} \perp_{\mathbf{w}} \text{span}(\Phi) \implies \tilde{\mathbf{q}}^H \mathbf{W} (\mathbf{F}^H (\mathbf{F}\mathbf{q} - \mathbf{p}) - d(\mathbf{w})\mathbf{q}) = 0,$$

giving $\left| \langle \mathbf{F}\tilde{\mathbf{q}}, \mathbf{u} \rangle_{\widetilde{\mathbf{W}}_\otimes} \right| = \frac{d(\mathbf{w})}{\gamma_\beta} \left| \tilde{\mathbf{q}}^H \mathbf{W} \mathbf{q} \right|$. By (7.8), the proof is complete. $\square$

7.2 Convergence for matrix-valued polynomial minimax approximations

In this subsection, we prove two interesting results of **m-d-Lawson** for the polynomial case (i.e., $d = 0$): (a) $d(\mathbf{w}^{(k+1)}) \geq d(\mathbf{w}^{(k)})$ for any $\beta \in [0, 2]$, and (b) the choice $\beta = 1$ is near-optimal in the sense that the lower bound in (7.6) attains its maximum when $\beta \geq 0$.

For these properties, we note that when $d = 0$, then $q \equiv 1$ implying that $\mathbf{q}^{(k+1)} \equiv \mathbf{q}^{(k)} \equiv \mathbf{1}_m \in \mathbb{R}^m$ and $(\mathbf{q}^{(k+1)})^H \mathbf{W}^{(k)} \mathbf{q}^{(k)} \equiv 1$ for all k . Thus the lower bound in (7.6) reduces to

$$\sqrt{d(\mathbf{w}^{(k+1)})} \geq d(\mathbf{w}^{(k)}) \frac{1}{\gamma_\beta^{(k)} \zeta_\beta^{(k)}}. \quad (7.9)$$

Following [48, Section 4], we define a near-optimal Lawson exponent $\beta_*^{(k)}$ as

$$\beta_*^{(k)} = \arg \min_{\beta \geq 0} \left(\gamma_\beta^{(k)} \zeta_\beta^{(k)} \right)^2. \quad (7.10)$$

Omitting the subscript k in each quantity related with the k th iteration as in the proof of Lemma 7.1, and recalling $\tau_\ell = \|F(x_\ell) - R(x_\ell)\|_F$ and $\mathcal{J}_{\tilde{\mathbf{w}}} = \{\ell : \tilde{w}_\ell \neq 0, 1 \leq \ell \leq m\} = \{\ell : w_\ell \tau_\ell \neq 0, 1 \leq \ell \leq m\}$ in Proposition 7.1, we obtain by $\tilde{w}_\ell = \frac{w_\ell \tau_\ell^\beta}{\gamma_\beta}$ ($1 \leq \ell \leq m$) and $q \equiv 1$ that

$$\begin{aligned} \nu(\beta) &:= \gamma_\beta^2 \zeta_\beta^2 = \gamma_\beta^2 \sum_{i=1}^s \sum_{j=1}^t \left(\sum_{\ell=1}^m \frac{\tilde{w}_\ell |f_{ij}(x_\ell) - p_{ij}(x_\ell)|^2}{\tau_\ell^{2\beta}} \right) \\ &= \gamma_\beta \sum_{\ell=1}^m \frac{w_\ell}{\tau_\ell^\beta} \underbrace{\left(\sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_\ell) - p_{ij}(x_\ell)|^2 \right)}_{=\tau_\ell^2} \\ &= \gamma_\beta \sum_{\ell=1}^m \frac{w_\ell \tau_\ell^2}{\tau_\ell^\beta} = \underbrace{\left(\sum_{\ell \in \mathcal{J}_{\tilde{\mathbf{w}}}} w_\ell \tau_\ell^\beta \right)}_{=\gamma_\beta} \left(\sum_{\ell \in \mathcal{J}_{\tilde{\mathbf{w}}}} w_\ell \tau_\ell^{2-\beta} \right). \end{aligned} \quad (7.11)$$

Note that $\nu(\beta)$ in (7.11) takes the same expression form as in [48]; particularly $\nu(0) = \nu(2) = d(\mathbf{w}^{(k)})$. Therefore, we can straightforwardly generalize [48, Proposition 4.2] to the case of matrix-valued polynomial minimax approximation:

Proposition 7.2. *For the matrix-valued polynomial minimax approximation, we have*

- (i) *the function $\nu(\beta)$ given in (7.11) is convex;*
- (ii) *$\beta = 1$ is the global minimizer of (7.10). In this sense, $\beta = 1$ achieves the maximum of the lower bound in (7.9) and can be viewed as the near-optimal Lawson exponent for m-d-Lawson in Algorithm 1;*
- (iii) *$\forall \beta \in [0, 2]$, the sequence $\{d(\mathbf{w}^{(k)})\}_k$ of m-d-Lawson satisfies $d(\mathbf{w}^{(k+1)}) \geq d(\mathbf{w}^{(k)})$.*

Proof. With $\nu(\beta)$ given in (7.11), the proof is finished according to [48, Section 4]. $\square$

7.3 Convergence for matrix-valued rational minimax approximations

We now analyze the convergence of m-d-Lawson (Algorithm 1) for general rational minimax approximation. Unlike the polynomial case, a key challenge arises because the numerator $|\langle \mathbf{q}^{(k+1)} \rangle^H \mathbf{W}^{(k)} \mathbf{q}^{(k)}|$ in the lower bound (7.6) also depends on the Lawson exponent β . However, owing to the structural similarities in the minimax framework, particularly the setting with a common denominator $q(x) \in \mathbb{P}_d$, the analysis in [48, Section 5] can be extended to obtain the following result.

Theorem 7.2. *At the k th iteration of m-d-Lawson in Algorithm 1, for $\mathbf{w}^{(k)} \in \mathcal{S}$, assume $d(\mathbf{w}^{(k)})$ is a simple eigenvalue of the matrix pencil $(A_{\mathbf{w}^{(k)}}, B_{\mathbf{w}^{(k)}})$ given in (3.1). Assume (A1) and (A2). Suppose $\tau_\ell^{(k)} := \|F(x_\ell) - R^{(k)}(x_\ell)\|_F > 0$, $\forall \ell \in \mathcal{J}_{\mathbf{w}^{(k)}} := \{j : w_j^{(k)} \neq 0, 1 \leq j \leq m\}$. Then*

- (i) *$\exists \beta_0 > 0$ so that $\forall \beta \in (0, \beta_0)$, we have $d(\mathbf{w}^{(k+1)}) \geq d(\mathbf{w}^{(k)})$;*

(ii) for any sufficiently small $\beta > 0$,

$$d(\mathbf{w}^{(k+1)}) = d(\mathbf{w}^{(k)}) \implies w_\ell^{(k)} \tau_\ell^{(k)} (\tau_\ell^{(k)} - c) = 0, \forall 1 \leq \ell \leq m, \quad (7.12)$$

where $c = \sqrt{d(\mathbf{w}^{(k)})} \leq \sqrt{e(R^{(k)})} = \max_{1 \leq \ell \leq m} \tau_\ell^{(k)}$;
 (iii) for item (ii), if additionally, $c \geq \max_{\ell \notin \mathcal{J}_{\mathbf{w}^{(k)}}} \tau_\ell^{(k)}$, then $R^{(k)}(x)$ is the minimax approximant of (1.2).

Proof. See Appendix G. $\square$

Before presenting our numerical results of the method **m-d-Lawson** for solving (1.2), we mention a recent related progress [47] on this new topic of rational minimax approximations for matrix-valued functions. In particular, for continuous functions $F(x)$ on a set $\mathcal{B} \subseteq \mathbb{C}$ dense in itself, the existence of the minimax approximant of

$$\inf_{R \in \mathcal{R}_{(s,t)}} \max_{x \in \mathcal{B}} \|F(x) - R(x)\|_F^2$$

is proved, which extends Walsh's foundational scalar result [41]; moreover, for a general compact set $\mathcal{B}$, [47] characterizes the local and global minimax approximants by deriving primal/dual matrix-valued Kolmogorov criteria and a Rutan-type sufficient condition for global optimality. These results have close relations with optimality conditions in Proposition 3.1, Corollary 3.1, strong duality in Theorem 4.1, and provide a theoretical basis the method **m-d-Lawson**, which has been applied to solve nonlinear eigenvalue problems in [46].

8 Numerical experiments

To verify the numerical performance of the method **m-d-Lawson**² in Algorithm 1, we implement it in MATLAB R2024b in double precision and carry out numerical experiments on a 14-inch MacBook Pro with an M3 Pro chip and 18Gb memory. The following numerical examples are from [16, 38, 39, 52], which have been used to test the following methods for matrix-valued rational approximations: **v-AAA-Lawson**³ [52], the block-AAA [16], RKFIT [5], Loewner framework interpolation approach [2, 17, 29] and the vector fitting (VF) approach in [19]. Codes of all the latter four are available at https://github.com/nla-group/block_aaa. For further numerical evaluations and results of **m-d-Lawson**, we refer the reader to [46] for its applications in solving nonlinear eigenvalue problems.

Example 8.1. The first test problem is a toy example from [16]. The target is a symmetric matrix-valued rational function $F : \mathbb{C} \rightarrow \mathbb{C}^{2 \times 2}$ given by

$$F(x) = \begin{bmatrix} \frac{2}{x+1} & \frac{3-x}{x^2+x-5} \\ \frac{3-x}{x^2+x-5} & \frac{2+x^2}{x^3+3x^2-1} \end{bmatrix}.$$

²See <https://github.com/longzih/mdlawson-code> for the MATLAB code of **m-d-Lawson**.

³See <https://www.mathworks.cn/matlabcentral/fileexchange/174740-v-aaa-lawson> for the MATLAB code of **v-AAA-Lawson**.

It can be seen that all entries f_{ij} of F can be written as type (5, 6) rational functions $f_{ij} = p_{ij}/q$ with the denominator $q(x) = (x+1)(x^2+x-5)(x^3+3x^2-1) \in \mathbb{P}_6$. To get sample data from F , we use $m = 1000$ equally spaced samples of F in the interval $[1, 100]i$ in the imaginary axis to have $\{x_\ell, F(x_\ell)\}_{\ell=1}^m$. To see the numerical performance of the tested methods, we also introduce a noise term σ_ℓ for each $f_{ij}(x_\ell)$ at node x_ℓ

$$\hat{f}_{ij}(x_\ell) = f_{ij}(x_\ell) + \sigma_\ell, \quad i, j = 1, 2, \text{ and } \ell = 1, \dots, m,$$

where the real and imaginary part of σ_ℓ are drawn from a normal distribution with mean zero and standard deviation (noise level) ς .

We remark that the current implementations of block-AAA⁴, RKFIT⁵, Loewner approach and VF provide rational approximations r_{ij} of type (d, d) for f_{ij} , while **m-d-Lawson** and **v-AAA-Lawson** are able to compute r_{ij} with user-defined types. Therefore, we set the exact types (5, 6) for the latter two, while choose type (6, 6) and default settings for others.

For a computed approximation, accuracy is evaluated using both the root mean squared error (RMSE) in [16] given by (1.5) and the maximum error $e(R) = \max_{x_\ell \in \mathcal{X}} \|F(x_\ell) - R(x_\ell)\|_F^2$. In Table 8.1, we report the average RMSE, $e(R)$ and runtime (s) over 10 random tests. Note **m-d-Lawson(10)** and **v-AAA-Lawson(10)** mean that the maximal number of Lawson iterations is $\ell_{\max} = 10$ for **m-d-Lawson** and **v-AAA-Lawson**, respectively.

Table 8.1 The average RMSE, $\sqrt{e(R)}$ and runtime (s) over 10 random tests for Example 8.1

Method RMSE	m-d-Lawson(10) type (5,6)	v-AAA-Lawson(10) type (5,6)	block-AAA type (6,6)	RKFIT(10) type (6,6)	Loewner type (6,6)	VF type (6,6)
$\varsigma = 0$	1.0320e-15	4.6607e-15	2.9514e-15	1.5406e-14	2.1706e-04	2.2637e-16
$\varsigma = 10^{-10}$	3.2917e-10	8.9433e-10	6.5689e-09	2.8139e-10	1.3767e-04	2.8238e-10
$\varsigma = 10^{-8}$	3.3440e-08	8.8972e-08	4.2916e-07	2.8151e-08	1.9399e-04	2.8240e-08
$\varsigma = 10^{-6}$	4.5715e-06	1.0650e-05	1.5047e-05	2.8350e-06	1.7105e-04	2.8431e-06
$\varsigma = 10^{-4}$	3.4177e-04	3.6645e-04	3.5141e-03	2.8227e-04	3.5490e-03	3.0333e-04
Method $\sqrt{e(R)}$	m-d-Lawson(10) type (5,6)	v-AAA-Lawson(10) type (5,6)	block-AAA type (6,6)	RKFIT(10) type (6,6)	Loewner type (6,6)	VF type (6,6)
$\varsigma = 0$	3.4414e-15	2.7243e-14	1.7939e-14	8.8405e-14	6.1212e-03	2.5466e-15
$\varsigma = 10^{-10}$	6.5239e-10	2.5588e-08	3.1151e-08	5.1673e-10	3.5601e-03	5.1934e-10
$\varsigma = 10^{-8}$	6.5416e-08	2.5463e-06	3.8708e-06	5.2108e-08	5.2838e-03	5.2149e-08
$\varsigma = 10^{-6}$	1.0512e-05	2.9845e-04	6.2092e-05	5.2671e-06	4.6763e-03	5.2826e-06
$\varsigma = 10^{-4}$	8.5338e-04	2.4750e-03	1.2549e-02	5.7604e-04	8.3391e-02	1.4158e-03
Method runtime (s)	m-d-Lawson(10) type (5,6)	v-AAA-Lawson(10) type (5,6)	block-AAA type (6,6)	RKFIT(10) type (6,6)	Loewner type (6,6)	VF type (6,6)
$\varsigma = 0$	2.1105e-02	1.2662e-01	2.8783e-02	5.1842e-02	1.5237e-01	1.4370e-02
$\varsigma = 10^{-10}$	1.9190e-02	7.5061e-02	2.3925e-02	4.8851e-02	1.7278e-01	1.4464e-02
$\varsigma = 10^{-8}$	1.9141e-02	9.0897e-02	2.4103e-02	5.0181e-02	1.7721e-01	1.4551e-02
$\varsigma = 10^{-6}$	1.9775e-02	8.9523e-02	2.4469e-02	5.1300e-02	1.8982e-01	1.4321e-02
$\varsigma = 10^{-4}$	1.9850e-02	1.0008e-01	2.4226e-02	4.9750e-02	1.6280e-01	1.4611e-02

⁴We notice that the order d of the denominator specified in block-AAA for approximating a matrix-valued function $F: \mathbb{C} \rightarrow \mathbb{C}^{n \times n}$ generally produces rational approximations of type (dn, dn) rational functions r_{ij} . Therefore, in our case with $q \in \mathbb{P}_6$, we specify the order 3 in block-AAA.

⁵RKFIT is an iterative method, and RKFIT(k) refers to RKFIT with k iterations.

Example 8.2. Our second test problem is a symmetric matrix-valued function $F : \mathbb{C} \rightarrow \mathbb{C}^{2 \times 2}$ from [16]

$$F(x) = \begin{bmatrix} \frac{x(1-2x \cot 2x)}{\tan x - x} + 10 & \frac{x(2x - \sin 2x)}{\sin(2x)(\tan x - x)} \\ \frac{x(2x - \sin 2x)}{\sin(2x)(\tan x - x)} & \frac{x(1-2x \cot 2x)}{\tan x - x} + 4 \end{bmatrix}.$$

This function is a 2-by-2 submatrix of the buckling example in [24], arising from the nonlinear eigenvalue problem in a buckling plate model. As each $f_{ij}(x)$ is not rational functions, we approximate f_{ij} by a type (d, d) rational function r_{ij} with $d = 10$ and with a common denominator. In the same settings for the noise as Example 8.1, we choose $m = 500$ logarithmically spaced sampling points in the interval $[10^{-2}, 10]i$, and report the accuracy of each method in Table 8.2.

Table 8.2 The average RMSE, $\sqrt{e(R)}$ and runtime (s) over 10 random tests for Example 8.2 with $d = 10$

Method RMSE	m-d-Lawson(10) type (10,10)	v-AAA-Lawson(10) type (10,10)	block-AAA type (10,10)	RKFIT(10) type (10,10)	Loewner type (10,10)	VF type (10,10)
$\varsigma = 0$	4.2986e-10	7.6848e-10	1.4303e-10	2.9241e-10	9.1307e-05	1.4763e-06
$\varsigma = 10^{-10}$	4.9808e-10	8.0165e-10	1.3108e-08	4.0177e-10	9.6323e-05	1.4763e-06
$\varsigma = 10^{-8}$	3.2636e-08	1.1411e-07	4.1369e-07	2.8176e-08	9.9527e-04	1.4763e-06
$\varsigma = 10^{-6}$	4.5794e-06	1.1822e-05	7.3229e-05	2.8382e-06	1.2183e-02	3.1778e-06
$\varsigma = 10^{-4}$	4.3981e-04	9.3440e-04	5.9198e-03	2.9181e-04	4.9153e+00	2.8811e-04
Method $\sqrt{e(R)}$	m-d-Lawson(10) type (10,10)	v-AAA-Lawson(10) type (10,10)	block-AAA type (10,10)	RKFIT(10) type (10,10)	Loewner type (10,10)	VF type (10,10)
$\varsigma = 0$	6.3915e-10	1.4332e-09	2.8392e-10	1.7800e-09	1.2674e-03	1.0515e-05
$\varsigma = 10^{-10}$	8.2054e-10	1.6173e-09	1.5136e-07	1.7630e-09	1.3620e-03	1.0515e-05
$\varsigma = 10^{-8}$	5.5735e-08	2.8724e-07	3.2453e-06	5.2552e-08	1.3289e-02	1.0512e-05
$\varsigma = 10^{-6}$	8.9376e-06	3.0720e-05	6.9040e-04	5.2238e-06	1.3572e-01	1.1117e-05
$\varsigma = 10^{-4}$	1.0781e-03	2.1639e-03	5.4932e-02	5.5378e-04	4.7214e+01	4.9630e-04
Method runtime (s)	m-d-Lawson(10) type (10,10)	v-AAA-Lawson(10) type (10,10)	block-AAA type (10,10)	RKFIT(10) type (10,10)	Loewner type (10,10)	VF type (10,10)
$\varsigma = 0$	2.3879e-02	1.8259e-02	2.5882e-02	5.9520e-02	5.6481e-02	2.4279e-02
$\varsigma = 10^{-10}$	1.9450e-02	1.3917e-02	2.2699e-02	5.1080e-02	4.5227e-02	1.3077e-02
$\varsigma = 10^{-8}$	1.9344e-02	1.5003e-02	2.2696e-02	5.1675e-02	4.3987e-02	1.2595e-02
$\varsigma = 10^{-6}$	1.9111e-02	1.3260e-02	2.2367e-02	5.2499e-02	4.0852e-02	1.4727e-02
$\varsigma = 10^{-4}$	1.9491e-02	1.3000e-02	2.2413e-02	5.1919e-02	4.0068e-02	1.2920e-02

With this example, we next demonstrate the property of complementary slackness (5.2) tied to optimality at extremal points. To this end, for type (6,6), in Figure 8.1, we plot the error $e_{ij}(x_\ell) = |f_{ij}(x_\ell) - r_{ij}(x_\ell)|$ for the individual r_{ij} , the F-norm error $\|F(x_\ell) - R(x_\ell)\|_F$ as well as the sequences of $\{\sqrt{e(R^{(k)})}\}_k$ and $\{\sqrt{d(\mathbf{w}^{(k)})}\}_k$. Noteworthy observations from the bottom-right subfigure of Figure 8.1 are that $\{\sqrt{d(\mathbf{w}^{(k)})}\}_k$ increases monotonically and the duality gap $\sqrt{e(R^{(k)})} - \sqrt{d(\mathbf{w}^{(k)})}$ converges to zero as the iteration count k increases. Furthermore, the top-right subfigure in Figure 8.1 reveals another interesting observation: exactly 11 extreme points attain the maximum approximation error $e(R)$ and the complementary slackness (5.2)

property is demonstrated numerically as

$$\max_{1 \leq \ell \leq m} |w_\ell(e(R) - \|F(x_\ell) - R(x_\ell)\|_F^2)| = 6.6680 \times 10^{-14}.$$

This aligns with a key property of minimax approximation which is related to the famous equioscillation property [40, Theorem 24.1] for the real scalar case. In contrast, such extreme points are absent in the other methods compared in Figure 8.2.

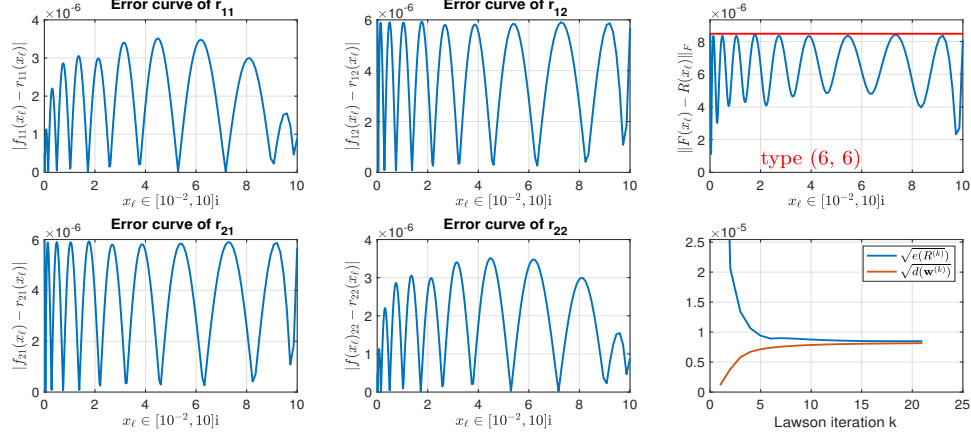


Fig. 8.1 The errors of the approximation $R(x)$ with type (6,6) computed by **m-d-Lawson(20)** for Example 8.2. The left two columns show the absolute errors $e_{ij}(x_\ell) = |f_{ij}(x_\ell) - r_{ij}(x_\ell)|$ of $r_{ij}(x)$, the up-right represents the F-norm error $\|F(x_\ell) - R(x_\ell)\|_F$ at each x_ℓ and the bottom-right demonstrates that the duality gap $\sqrt{e(R^{(k)})} - \sqrt{d(w^{(k)})}$ decreases as k increases.

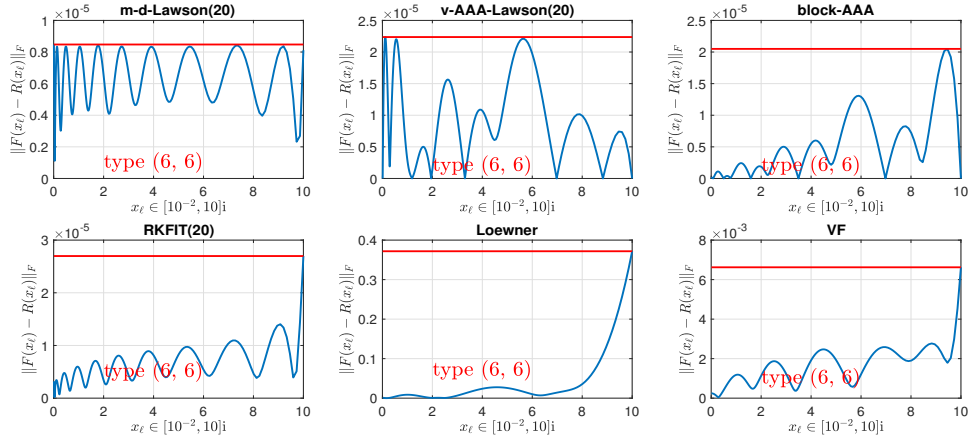


Fig. 8.2 The F-norm errors $\|F(x_\ell) - R(x_\ell)\|_F$ of the approximation $R(x)$ with type (6,6) computed by different methods for Example 8.2.

We also use this example to demonstrate the performance of **m-d-Lawson** for computing the best matrix-valued *polynomial* approximation of $F(x)$. In particular, we set the degree $d = 0$ for the denominator and $n_{ij} = 12$ for all numerators; thus $R(x)$ becomes a matrix-valued polynomial

$$P(x) = \begin{bmatrix} p_{11}(x) & p_{12}(x) \\ p_{21}(x) & p_{22}(x) \end{bmatrix} : \mathbb{C} \rightarrow \mathbb{C}^{2 \times 2}, \quad p_{ij} \in \mathbb{P}_{12}.$$

In this case, similarly, we report the error $e_{ij}(x_\ell) = |f_{ij}(x_\ell) - p_{ij}(x_\ell)|$, $\|F(x_\ell) - P(x_\ell)\|_F$ as well as the sequences of $\{\sqrt{e(P^{(k)})}\}_k$ and $\{\sqrt{d(\mathbf{w}^{(k)})}\}_k$ in Figure 8.3. Comparing with Figure 8.1, we observe that the polynomial and rational cases share a similar performance with

$$\max_{1 \leq \ell \leq m} |w_\ell(e(R) - \|F(x_\ell) - R(x_\ell)\|_F^2)| = 2.2900 \times 10^{-11}.$$

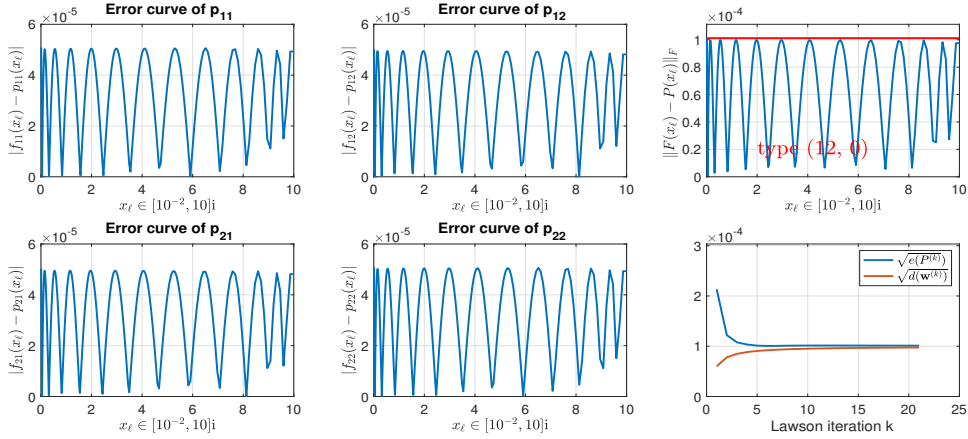


Fig. 8.3 The errors of the polynomial approximation $P(x)$ with type (12,0) computed by **m-d-Lawson(20)** for Example 8.2. The left two columns show the absolute errors $e_{ij}(x_\ell) = |f_{ij}(x_\ell) - p_{ij}(x_\ell)|$ of $p_{ij}(x)$, the up-right represents the F-norm error $\|F(x_\ell) - P(x_\ell)\|_F$ at each x_ℓ , and the bottom-right demonstrates that the duality gap $\sqrt{e(P^{(k)})} - \sqrt{d(\mathbf{w}^{(k)})}$ decreases as k increases.

A final illustration with Example 8.2 is to demonstrate the errors of RMSE and $e(R)$ with various types other than (d, d) . In particular, we apply **v-AAA-Lawson(20)** and **m-d-Lawson(20)** to two cases: $(n_{ij}, 20)$ and $(n_{ij}, n_{ij} + 2)$ with $n_{ij} = 1, \dots, 20$. Figure 8.4 illustrates RMSE and $e(R)$ for these cases, which implies that **m-d-Lawson(20)** generally produces more accurate approximation than **v-AAA-Lawson(20)**.

Example 8.3. We now test the case arising from the designed duplexer device with given poles and zeros in [38, 39]. In this example, three rational functions, i.e.,

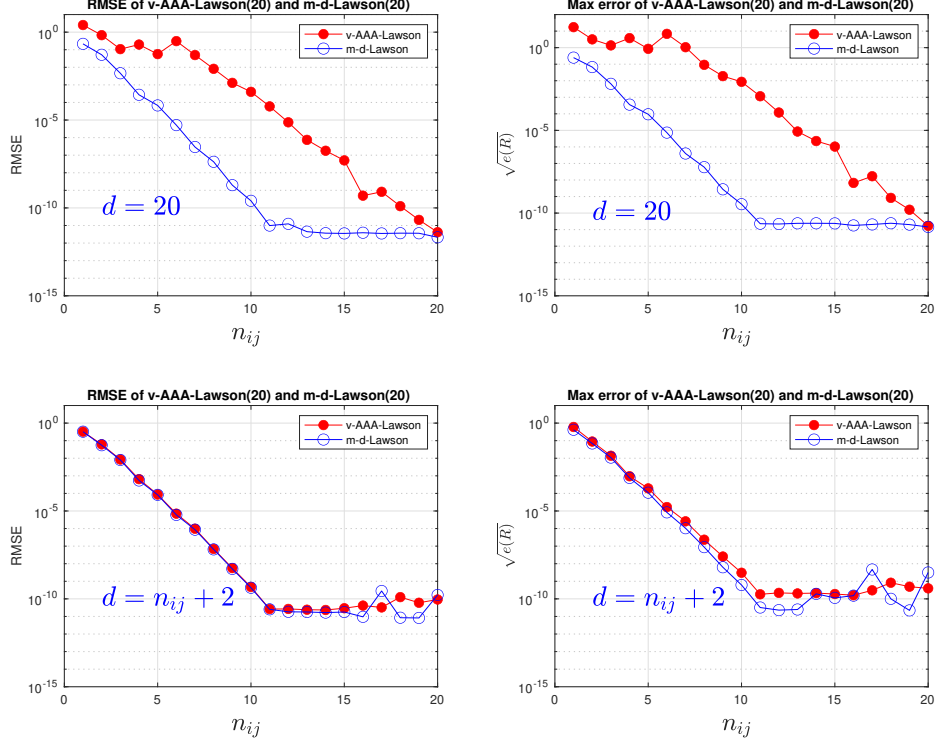


Fig. 8.4 RMSE (left column) and $\sqrt{e(R)}$ (right column) from v-AAA-Lawson(20) and m-d-Lawson(20) based on rational approximation types $(n_{ij}, 20)$ and $(n_{ij} + 2, n_{ij})$ with $n_{ij} = 2, \dots, 20$ for Example 8.2.

scattering parameters,

$$s_{11}(x) = \frac{p_N(x)}{p_D(x)}, \quad s_{21}(x) = \frac{p_T(x)}{p_D(x)}, \quad \text{and} \quad s_{31}(x) = \frac{p_R(x)}{p_D(x)}, \quad (8.1)$$

need to be reconstructed from sampled responses measured at a set of frequencies. Here, $p_N \in \mathbb{P}_{20}$, $p_D \in \mathbb{P}_{20}$, $p_T \in \mathbb{P}_{12}$, $p_R \in \mathbb{P}_{12}$ are complex polynomials, and the sample nodes are from the line $x = i\omega$ with $\omega \in \mathbb{R}$ being the frequency. In our application, there are 401 (i.e., $m = 400$) equidistant sampled frequencies $\mathcal{X} = \{(-2 + \frac{\ell}{400})i\}_{\ell=0}^{400}$ as well as responses $\{s_{j1}(x_\ell)\}_{\ell=0}^{400}$ for $j = 1, 2, 3$. The zeros of associated polynomials of $p_D(x)$, $p_N(x)$, $p_T(x)$ and $p_R(x)$ are given in Table 8.3.

To apply v-AAA-Lawson and m-d-Lawson, we set $s = 3$ and $t = 1$ to form a vector-valued complex function $F(x) = [s_{11}(x), s_{21}(x), s_{31}(x)]^T : \mathbb{C} \rightarrow \mathbb{C}^3$. In order to verify the robustness, we put a random perturbation to the responses as follows

$$\hat{s}_{j1}(x_\ell) = s_{j1}(x_\ell) + \sigma_\ell, \quad j = 1, 2, 3, \quad \text{and} \quad \ell = 0, \dots, m,$$

where σ_ℓ is the noise with noise level ς described in Example 8.1. In the right column of Figure 8.5, we plot the fitting curves $y_j(x) = 20 \log_{10} |s_{j1}(x)|$ ($j = 1, 2, 3$) as well as their rational reconstructions computed from **v-AAA-Lawson**(10) and **m-d-Lawson**(10) for the noise level $\varsigma = 10^{-8}$; the left two columns plot the roots (solid points) given in Table 8.3 and the computed ones (circles) from **v-AAA-Lawson**(10) and **m-d-Lawson**(10).

Table 8.3 Roots of the associated polynomials of a duplexer from [38, 39]

roots of $p_D(x)$	roots of $p_N(x)$	roots of $p_T(x)$	roots of $p_R(x)$
-0.0396-1.0564i	-0.0050-0.0173i	0.0130-0.0070i	-0.0071+0.1128i
-0.1099-0.9971i	0.0404+0.0037i	-0.0048+0.0181i	-0.0228+0.1335i
-0.1452-0.8985i	-0.0221-0.0921i	-0.0309-0.0016i	0.0063+0.1481i
-0.0349+1.0418i	-0.0009+0.1261i	-0.0098-0.0443i	-0.0461+0.1752i
-0.0990+0.9864i	-0.0151+0.1542i	-0.0501-0.0915i	-0.0068+0.2229i
-0.1481-0.7199i	-0.0217-0.2031i	-0.0647-0.2039i	-0.0740+0.2899i
-0.1404+0.8811i	0.0231+0.2331i	-0.0994-0.3527i	-0.0735+0.4291i
-0.1637+0.7315i	-0.0223+0.2925i	-0.0890-0.5222i	-0.0954+0.5850i
-0.1497-0.5206i	-0.0621-0.3582i	-0.0858-0.7178i	-0.0809+0.7525i
-0.1668+0.5697i	0.0310+0.4292i	-0.0650-0.9007i	-0.0765+0.8988i
-0.1388-0.3422i	-0.0142-0.5192i	-0.0527-1.0230i	-0.0430+1.0096i
-0.1484+0.4140i	-0.0309+0.5772i	-0.0072-1.1030i	-0.0053+1.1038i
-0.1170+0.2846i	-0.0249-0.7174i		
-0.0999-0.1895i	0.0352+0.7185i		
-0.0767+0.1975i	-0.0281+0.8746i		
-0.0386+0.1414i	0.0049-0.9350i		
-0.0111+0.1166i	0.0957-0.9407i		
-0.0636-0.0772i	0.0497+0.9520i		
-0.0427+0.0014i	-0.0163+1.0071i		
-0.0200-0.0024i	-0.0299-1.0232i		

9 Conclusions

Rational approximation of matrix-valued functions plays an important role in various applications, including nonlinear eigenvalue problems [22]. Although several numerical methods based on discrete point sets have been proposed, most lack a well-defined optimization objective, resulting in approximations without guaranteed optimality.

In this paper, inspired by classical minimax approximation for scalar functions, we rigorously formulate a matrix-valued rational minimax approximation problem (1.2). Exploiting recent advances in duality-based minimax approximation [45, 48, 50], we reformulate (1.2) as a max-min dual problem, which yields several key advantages:

- Computational tractability: The dual problem reduces to a real-valued optimization over the probability simplex, allowing us to extend the **d-Lawson** iteration [50] to the proposed **m-d-Lawson** method.
- Convergence guarantees: We establish the convergence of **m-d-Lawson** for both matrix-valued polynomial and rational approximations, rigorously analyzing the influence of the Lawson exponent β .
- Theoretical insights: Our duality analysis provides a sufficient condition (Theorem 4.1) ensuring that the dual solution coincides with the best approximant for (1.2).

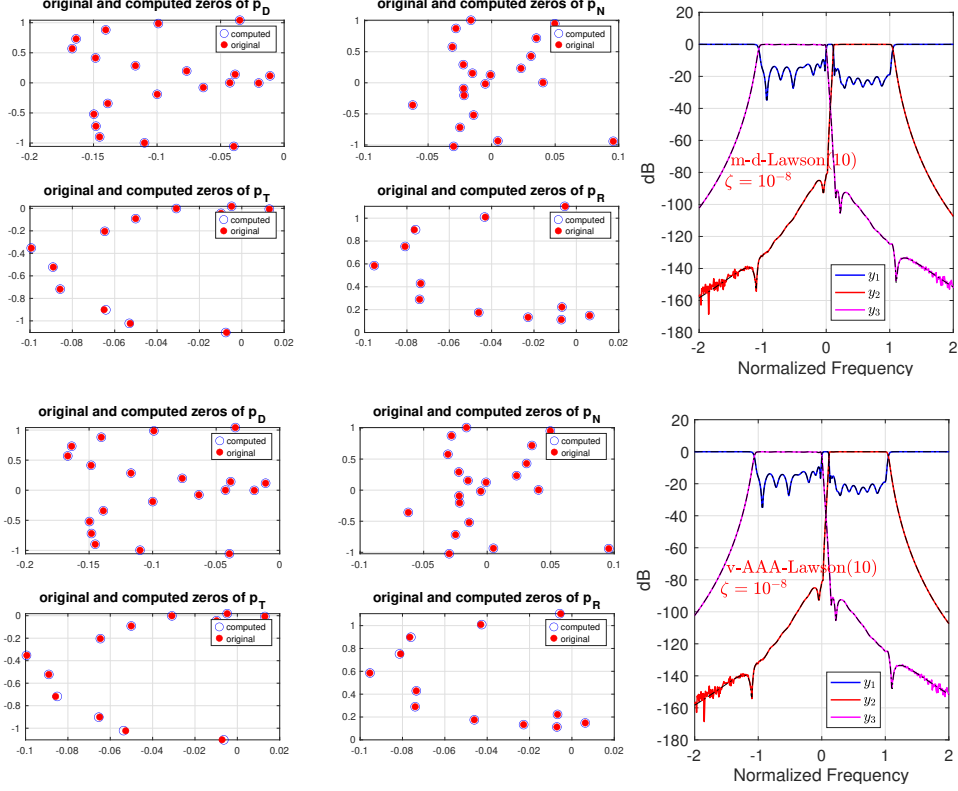


Fig. 8.5 Synthesized $y_j(x)$ (thick line in the right column) for $j = 1, 2, 3$ and fitted responses (thin dashed line in the right column) from **m-d-Lawson(10)** (top) and **v-AAA-Lawson(10)** (bottom) with noise level $\zeta = 10^{-8}$. The left two columns plot the roots (solid points) given in Table 8.3 and the computed ones (circles).

Furthermore, we characterize optimal solutions via complementary slackness for extreme points and strong duality.

Numerical experiments demonstrate that **m-d-Lawson** can solve (1.2) efficiently, offering a new computational framework for matrix-valued rational approximation.

A Appendix

Proof of Theorem 2.1. For any $0 \neq q \in \mathbb{P}_d$, as $m \geq d + 2$, there exists $x_j \in \mathcal{X}$ with $q(x_j) \neq 0$. This implies that $\eta_2 \geq 0$. Moreover, $\eta_2 < \infty$ as we can choose $p_{ij} \equiv 0$ ($1 \leq i \leq s, 1 \leq j \leq t$) and $q \equiv 1$ to bound the left hand side of each constraint of (2.6) by a constant.

Let $\{(\eta^{(k)}, \{p_{ij}^{(k)}\}, q^{(k)})\}_k$ be a feasible sequence such that $p_{ij}^{(k)} \in \mathbb{P}_{n_{ij}}, 0 \neq q^{(k)} \in \mathbb{P}_d$ and $\eta^{(k)} \rightarrow \eta_2$ as $k \rightarrow \infty$. For any $\varsigma^{(k)} \neq 0$, since $\{(\eta^{(k)}, \varsigma^{(k)}\{p_{ij}^{(k)}\}, \varsigma^{(k)}q^{(k)})\}$ is also feasible for (2.6), $\varsigma^{(k)}$ can be chosen so that the coefficient vector $\mathbf{b}^{(k)} = [b_0^{(k)}, \dots, b_d^{(k)}]^T \in$

$\mathbb{C}^{d+1}$ of $\varsigma^{(k)}q^{(k)}(x) = \sum_{i=0}^d b_i^{(k)}x^i$ satisfies $\|\mathbf{b}^{(k)}\|_2 = 1$ ($\forall k \geq 0$). Let $\tilde{p}_{ij}^{(k)} := \varsigma^{(k)}p_{ij}^{(k)}$ and $\tilde{q}^{(k)} := \varsigma^{(k)}q^{(k)}$. Thus $\{\tilde{q}^{(k)}\}_k$ has a convergent subsequence resulting from a convergent subsequence of $\{\mathbf{b}^{(k)}\}_k$. For simplicity of presentation, we assume that $\{\tilde{q}^{(k)}\}_k$ itself converges to $0 \neq q \in \mathbb{P}_d$, and $\eta^{(k)} \leq \rho_0$ ($\forall k \geq 0$) for some $\rho_0 > 0$.

We next show that for each pair (i, j) , $\{p_{ij}^{(k)}\}_k$ also contains a convergent subsequence. Indeed, by

$$|f_{ij}(x_\ell)\tilde{q}^{(k)}(x_\ell) - \tilde{p}_{ij}^{(k)}(x_\ell)|^2 \leq \sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_\ell)\tilde{q}^{(k)}(x_\ell) - \tilde{p}_{ij}^{(k)}(x_\ell)|^2 \leq \eta^{(k)}|\tilde{q}^{(k)}(x_\ell)|^2,$$

we have

$$|\tilde{p}_{ij}^{(k)}(x_\ell)| \leq \sqrt{\eta^{(k)}}|\tilde{q}^{(k)}(x_\ell)| + |f_{ij}(x_\ell)\tilde{q}^{(k)}(x_\ell)| \leq (\sqrt{\rho_0} + f_{\max})\rho_1 =: \tilde{\rho}_1, \quad \forall x_\ell \in \mathcal{X},$$

where $\rho_1 = \sum_{i=0}^d |x_J|^i$ with $|x_J| = \max_{1 \leq j \leq m} |x_j|$. Denoting $\tilde{p}_{ij}^{(k)}(x) = \sum_{h=0}^{n_{ij}} a_h^{(k)} x^h$ in the monomial basis, we can choose the first $n_{ij} + 1$ nodes $\{x_\ell\}_{\ell=1}^{n_{ij}+1}$ to have

$$[\tilde{p}_{ij}^{(k)}(x_1), \dots, \tilde{p}_{ij}^{(k)}(x_{n_{ij}+1})]^T = V(x_1, \dots, x_{n_{ij}+1})[a_0^{(k)}, \dots, a_{n_{ij}}^{(k)}]^T,$$

where $V(x_1, \dots, x_{n_{ij}+1})$ is the Vandermonde matrix associated with $\{x_\ell\}_{\ell=1}^{n_{ij}+1}$. This gives

$$\|[a_0^{(k)}, \dots, a_{n_{ij}}^{(k)}]\|_2 \leq \tilde{\rho}_1 \sqrt{n_{ij} + 1} \| [V(x_1, \dots, x_{n_{ij}+1})]^{-1} \|_2, \quad \forall k \geq 0.$$

Therefore, $\{\tilde{p}_{ij}^{(k)}\}_k$ also has a convergent subsequence with a limit polynomial $p_{ij} \in \mathbb{P}_{n_{ij}}$ resulting from a convergent subsequence of $\{[a_0^{(k)}, \dots, a_{n_{ij}}^{(k)}]\}_k$. Consequently, we can find $\{\eta_2, \{p_{ij}\}, q\}$ with $0 \neq q \in \mathbb{P}_d$ which solves (2.6). $\square$

B Appendix

Proof of Theorem 2.2. The proof is adopted by a slight modification of [48, Theorem 2.4]. First, $\eta_2 \leq \eta_\infty$ holds due to the fact that for sequence $\{\eta_\infty^{(k)}, R^{(k)}\}_k$, $\forall R^{(k)} \in \mathcal{R}_{(s,t)}$ with $q^{(k)}(x) \neq 0$ ($\forall x \in \mathcal{X}$), $(\eta_\infty^{(k)}, \{p_{ij}^{(k)}\}, q^{(k)})$ is feasible for (2.6), and thus $\eta_2 \leq \eta_\infty^{(k)} \rightarrow \eta_\infty$.

We prove $\eta_2 = \eta_\infty$ by ruling out the possibility $\eta_2 < \eta_\infty$. Assume $\eta_2 < \eta_\infty$. By Theorem 2.1, we let $(\eta_2, \{\hat{p}_{ij}\}, \hat{q})$ be a solution to (2.6) where $\hat{p}_{ij} \in \mathbb{P}_{n_{ij}}$ ($1 \leq i \leq s, 1 \leq j \leq t$) and $0 \neq \hat{q} \in \mathbb{P}_d$. Suppose, without loss of generality, that $\hat{q}(x_\ell) = 0$ for $\ell = 1, \dots, h$. As $\hat{q} \neq 0$, we have $h \leq d$. For any pair (i, j) , we consider the constraint of (2.6) on nodes x_ℓ , $\forall \ell = 1, 2, \dots, h$. Noting from

$$\sum_{i=1}^s \sum_{j=1}^t |f_{ij}(\ell)\hat{q}(x_\ell) - \hat{p}_{ij}(x_\ell)|^2 = \sum_{i=1}^s \sum_{j=1}^t |\hat{p}_{ij}(x_\ell)|^2 \leq \eta_2 \hat{q}(x_\ell) = 0,$$

we have $\hat{p}_{ij}(x_\ell) = 0, \forall \ell = 1, \dots, h$. If $h = 0$ (i.e., $\hat{q}(x_\ell) \neq 0 \forall \ell = 1, 2, \dots, m$), then the proof is finished as

$$\sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_\ell) \hat{q}(x_\ell) - \hat{p}_{ij}(x_\ell)|^2 \leq \eta_2 |\hat{q}(x_\ell)|^2 \implies \sum_{i=1}^s \sum_{j=1}^t \left| f_{ij}(x_\ell) - \frac{\hat{p}_{ij}(x_\ell)}{\hat{q}(x_\ell)} \right|^2 \leq \eta_2 < \eta_\infty,$$

contradicting with the fact that η_∞ is the infimum of (1.2).

Thus, we next consider the case $h \geq 1$. We will show that there is a parameterized form $\hat{R}(x; \delta) \in \mathcal{R}_{(s,t)}$ with its (i, j) th component $\hat{r}_{ij}(x; \delta)$ given by

$$\hat{r}_{ij}(x; \delta) = \frac{\hat{p}_{ij}(x; \delta)}{\hat{q}(x; \delta)} := \frac{\hat{p}_{ij}(x) + \delta \cdot p_{ij}(x)}{\hat{q}(x) + \delta \cdot q(x)}$$

for some polynomials $p_{ij} \in \mathbb{P}_{n_{ij}}$ ($1 \leq i \leq s, 1 \leq j \leq t$) and $q \in \mathbb{P}_d$ so that

$$\max_{1 \leq \ell \leq m} \sum_{i=1}^s \sum_{j=1}^t \left| f_{ij}(x_\ell) - \frac{\hat{p}_{ij}(x_\ell; \delta)}{\hat{q}(x_\ell; \delta)} \right|^2 < \eta_\infty. \quad (\text{B.1})$$

This implies that η_∞ is not the infimum of (1.2), a contradiction.

To this end, choose $q \equiv 1$. Since $h \leq d \leq \min_{1 \leq i \leq s, 1 \leq j \leq t} \{n_{ij}\}$, there are interpolation polynomials $p_{ij} \in \mathbb{P}_{n_{ij}}$ fulfilling conditions $p_{ij}(x_\ell) = f_{ij}(x_\ell)$ ($1 \leq \ell \leq h$); hence $\left| f_{ij}(x_\ell) - \frac{p_{ij}(x_\ell)}{q(x_\ell)} \right| = 0, 1 \leq \ell \leq h$, leading to

$$\sum_{i=1}^s \sum_{j=1}^t \left| f_{ij}(x_\ell) - \frac{p_{ij}(x_\ell)}{q(x_\ell)} \right|^2 = 0$$

and $q(x_\ell) \neq 0, \forall \ell = 1, \dots, h$.

On the other hand, for any sufficiently small $\delta \neq 0$, we know that $\hat{q}(x_\ell; \delta) = \hat{q}(x_\ell) + \delta q(x_\ell) = \hat{q}(x_\ell) + \delta \neq 0, 1 \leq \ell \leq m$. Moreover, the parameterized $\frac{\hat{p}_{ij}(x; \delta)}{\hat{q}(x; \delta)}$ satisfies

$$\begin{aligned} \sum_{i=1}^s \sum_{j=1}^t \left| \frac{\hat{p}_{ij}(x_\ell; \delta)}{\hat{q}(x_\ell; \delta)} - f_{ij}(x_\ell) \right|^2 &= \sum_{i=1}^s \sum_{j=1}^t \left| \frac{\hat{p}_{ij}(x_\ell) + \delta \cdot p_{ij}(x_\ell)}{\hat{q}(x_\ell) + \delta \cdot q(x_\ell)} - f_{ij}(x_\ell) \right|^2 \\ &= \sum_{i=1}^s \sum_{j=1}^t \left| \frac{p_{ij}(x_\ell)}{q(x_\ell)} - f_{ij}(x_\ell) \right|^2 = 0 < \eta_\infty, \forall \ell = 1, \dots, h. \end{aligned}$$

Furthermore, $\forall \ell = h+1, \dots, m$, we have

$$\sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_\ell) \hat{q}(x_\ell) - \hat{p}_{ij}(x_\ell)|^2 \leq \eta_2 |\hat{q}(x_\ell)|^2 \implies \sum_{i=1}^s \sum_{j=1}^t \left| f_{ij}(x_\ell) - \frac{\hat{p}_{ij}(x_\ell)}{\hat{q}(x_\ell)} \right|^2 \leq \eta_2 < \eta_\infty,$$

and for each pair (i, j) ,

$$\left| \frac{\hat{p}_{ij}(x_\ell) + \delta \cdot p_{ij}(x_\ell)}{\hat{q}(x_\ell) + \delta \cdot q(x_\ell)} - \frac{\hat{p}_{ij}(x_\ell)}{\hat{q}(x_\ell)} \right| = |\delta| \cdot \left| \frac{p_{ij}(x_\ell)\hat{q}(x_\ell) - \hat{p}_{ij}(x_\ell)q(x_\ell)}{(\hat{q}(x_\ell) + \delta)q(x_\ell)} \right| \rightarrow 0, \quad \text{as } \delta \rightarrow 0.$$

Consequently, for any sufficiently small δ , it follows

$$\begin{aligned} \sum_{i=1}^s \sum_{j=1}^t \left| \frac{\hat{p}_{ij}(x_\ell; \delta)}{\hat{q}(x_\ell; \delta)} - f_{ij}(x_\ell) \right|^2 &\leq \sum_{i=1}^s \sum_{j=1}^t \left(\left| \frac{\hat{p}_{ij}(x_\ell; \delta)}{\hat{q}(x_\ell; \delta)} - \frac{\hat{p}_{ij}(x_\ell)}{\hat{q}(x_\ell)} \right| + \left| f_{ij}(x_\ell) - \frac{\hat{p}_{ij}(x_\ell)}{\hat{q}(x_\ell)} \right| \right)^2 \\ &< \eta_\infty, \quad \forall \ell = h+1, \dots, m. \end{aligned}$$

This leads to (B.1), and we have $\eta_2 = \eta_\infty$ by contradiction.

For the last part, according to $\eta_2 = \eta_\infty$ and Theorem 2.1, whenever (1.2) has a solution $R \in \mathcal{R}_{(s,t)}$ with $q(x_j) \neq 0$ ($1 \leq j \leq m$), the tuple $(\eta_\infty, \{p_{ij}\}, q)$ is a solution of (2.6). $\square$

C Appendix

Proof of Theorem 2.3. We first remark that, for any non-negative $\mathbf{w} \neq \mathbf{0}$, the minimization of (2.14) is reachable at a pair $(\mathbf{a}, \mathbf{b})$ as it is a trace minimization for a Hermitian positive semi-definite pencil, and the solution can be obtained by [27, Theorems 2.1 and 4.1] (more detailed information on computation of $(\mathbf{a}, \mathbf{b})$ will be given in Section 3). Let $(\{\tilde{p}_{ij}\}, \tilde{q})$ for $1 \leq i \leq s$, $1 \leq j \leq t$ be the corresponding polynomials at the minimizer of (2.14).

We next show that $d(\mathbf{w}) \leq \eta_\infty, \forall \mathbf{w} \geq \mathbf{0}$, which then guarantees $\sup_{\mathbf{w} \geq \mathbf{0}} d(\mathbf{w}) \leq \eta_\infty$. Because η_∞ is the infimum in (1.2), for any $\epsilon > 0$, there is an $R(x) \in \mathcal{R}_{(s,t)}$ with $q(x_j) \neq 0$ ($\forall x_\ell \in \mathcal{X}$) so that

$$\max_{1 \leq \ell \leq m} \|F(x_\ell) - R(x_\ell)\|_{\mathbb{F}}^2 \leq \eta_\infty + \epsilon$$

which gives

$$\sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_\ell) - r_{ij}(x_\ell)|^2 \leq \eta_\infty + \epsilon, \quad 1 \leq \ell \leq m. \quad (\text{C.1})$$

As $\mathbf{w} \neq \mathbf{0}$ and $q(x_\ell) \neq 0$ ($\forall x_\ell \in \mathcal{X}$), we can choose a scalar $\tau \neq 0$ so that $\hat{q} = \tau q$, $\hat{p}_{ij} = \tau p_{ij}$ for $1 \leq i \leq s$, $1 \leq j \leq t$ satisfying $\sum_{\ell=1}^m w_\ell |\hat{q}(x_\ell)|^2 = 1$. Thus, as $(\{\tilde{p}_{ij}\}, \tilde{q})$ for $1 \leq i \leq s$, $1 \leq j \leq t$ are the corresponding polynomials for (2.14) at $\mathbf{w}$, it holds that

$$\begin{aligned} d(\mathbf{w}) &= \sum_{\ell=1}^m w_\ell \left(\sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_\ell)\tilde{q}(x_\ell) - \tilde{p}_{ij}(x_\ell)|^2 \right) \\ &\leq \sum_{\ell=1}^m w_\ell \left(\sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_\ell)\hat{q}(x_\ell) - \hat{p}_{ij}(x_\ell)|^2 \right) \quad (\text{because } (\{\tilde{p}_{ij}\}, \tilde{q}) \text{ are optimal}) \end{aligned}$$

$$\begin{aligned}
&= \sum_{\ell=1}^m w_\ell |\widehat{q}(x_\ell)|^2 \cdot \left(\sum_{i=1}^s \sum_{j=1}^t \left| f_{ij}(x_\ell) - \frac{\widehat{p}_{ij}(x_\ell)}{\widehat{q}(x_\ell)} \right|^2 \right) \\
&= \sum_{\ell=1}^m w_\ell |\widehat{q}(x_\ell)|^2 \cdot \left(\sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_\ell) - r_{ij}(x_\ell)|^2 \right) \\
&\leq \sum_{\ell=1}^m w_\ell |\widehat{q}(x_\ell)|^2 (\eta_\infty + \epsilon) = \eta_\infty + \epsilon,
\end{aligned}$$

where we have used (C.1) and $\sum_{\ell=1}^m w_\ell |\widehat{q}(x_\ell)|^2 = 1$. Letting $\epsilon \rightarrow 0$ gives the desired $d(\mathbf{w}) \leq \eta_\infty$.

Finally, to prove $\sup_{\mathbf{w} \geq \mathbf{0}} d(\mathbf{w}) = \max_{\mathbf{w} \in \mathcal{S}} d(\mathbf{w})$, we only need to show $\sup_{\mathbf{w} \geq \mathbf{0}} d(\mathbf{w}) \leq \max_{\mathbf{w} \in \mathcal{S}} d(\mathbf{w})$. For any $\mathbf{w} \geq \mathbf{0}$, let $\widetilde{\mathbf{w}} = \mathbf{w}_\varsigma \in \mathcal{S}$ with $\varsigma = \frac{1}{\mathbf{w}^\top \mathbf{e}}$ and set correspondingly $\widetilde{W} = \text{diag}(\widetilde{\mathbf{w}})$ and $\widetilde{W}_\otimes := I_g \otimes \widetilde{W}$. Note

$$\begin{aligned}
d(\widetilde{\mathbf{w}}) &= \min_{\substack{\mathbf{a} \in \mathbb{C}^n, \mathbf{b} \in \mathbb{C}^{d+1} \\ \|\sqrt{\widetilde{W}} \Phi \mathbf{b}\|_2 = 1}} \left\| \sqrt{\widetilde{W}_\otimes} [-\Theta, \mathbf{F} \Phi] \begin{bmatrix} \mathbf{a} \\ \mathbf{b} \end{bmatrix} \right\|_2^2 \\
&= \min_{\substack{\sqrt{\varsigma} \mathbf{a} \in \mathbb{C}^n, \sqrt{\varsigma} \mathbf{b} \in \mathbb{C}^{d+1} \\ \|\sqrt{W} \Phi(\sqrt{\varsigma} \mathbf{b})\|_2 = 1}} \left\| \sqrt{W_\otimes} [-\Theta, \mathbf{F} \Phi] \begin{bmatrix} \sqrt{\varsigma} \mathbf{a} \\ \sqrt{\varsigma} \mathbf{b} \end{bmatrix} \right\|_2^2 = d(\mathbf{w}),
\end{aligned}$$

and thus the proof is complete. $\square$

D Appendix

Proof of Proposition 3.2. For $\mathbf{w} > \mathbf{0}$, we know from Proposition 3.1 that $\widetilde{n}_1 = n$ and $\widetilde{n}_2 = d + 1$, and $d(\mathbf{w})$ is a simple eigenvalue of a positive semi-definite Hermitian matrix $S(\mathbf{w})$ in (3.4) and thus is differentiable at $\mathbf{w}$ (see e.g., [15, Section 7.2.2]). We obtain the formulation of the gradient $\nabla d(\mathbf{w})$ via computing the directional derivative along a given direction $\mathbf{h} \in \mathbb{R}^m$. To this end, let $\mathbf{w}(t) = \mathbf{w} + t\mathbf{h}$ and $\mathbf{c}(\mathbf{w}(t))$ be the corresponding eigenvector in (3.4) for any sufficiently small $t \in \mathbb{R}$. We have

$$(\mathbf{A}_{\mathbf{w}(t)} - d(\mathbf{w}(t))\mathbf{B}_{\mathbf{w}(t)})\mathbf{c}(\mathbf{w}(t)) = \mathbf{0}, \quad (\mathbf{c}(\mathbf{w}(t)))^\text{H} \mathbf{B}_{\mathbf{w}(t)} \mathbf{c}(\mathbf{w}(t)) = 1.$$

Differentiate the first term with respect to t and note $\dot{\mathbf{B}}_{\mathbf{w}(0)} = [0, \Phi]^\text{H} \text{diag}(\mathbf{h}) [0, \Phi]$ and $\dot{\mathbf{A}}_{\mathbf{w}(0)} = [-\Theta, \mathbf{F} \Phi]^\text{H} (I_g \otimes \text{diag}(\mathbf{h})) [-\Theta, \mathbf{F} \Phi]$ to have

$$(\dot{\mathbf{A}}_{\mathbf{w}(0)} - \dot{d}(\mathbf{w}(0))\mathbf{B}_{\mathbf{w}(0)} - d(\mathbf{w}(0))\dot{\mathbf{B}}_{\mathbf{w}(0)})\mathbf{c}(\mathbf{w}(0)) + (\mathbf{A}_{\mathbf{w}(0)} - d(\mathbf{w}(0))\mathbf{B}_{\mathbf{w}(0)})\dot{\mathbf{c}}(\mathbf{w}(0)) = \mathbf{0}.$$

Pre-multiplying $(\mathbf{c}(\mathbf{w}(0)))^\text{H}$ on both sides and using

$$(\mathbf{c}(\mathbf{w}(0)))^\text{H} (\mathbf{A}_{\mathbf{w}(0)} - d(\mathbf{w}(0))\mathbf{B}_{\mathbf{w}(0)}) = 0 \quad \text{and} \quad (\mathbf{c}(\mathbf{w}(t)))^\text{H} \mathbf{B}_{\mathbf{w}(t)} \mathbf{c}(\mathbf{w}(t)) = 1,$$

we get

$$\begin{aligned} (\nabla d(\mathbf{w}))^T \mathbf{h} &= \dot{d}(\mathbf{w}(0)) = (\mathbf{c}(\mathbf{w}(0)))^H \dot{A}_{\mathbf{w}(0)} \mathbf{c}(\mathbf{w}(0)) - d(\mathbf{w}(0)) (\mathbf{c}(\mathbf{w}(0)))^H \dot{B}_{\mathbf{w}(0)} \mathbf{c}(\mathbf{w}(0)) \\ &= \left(\sum_{i=1}^s \sum_{j=1}^t |F_{ij} \mathbf{q} - \mathbf{p}_{ij}|^2 - d(\mathbf{w}) |\mathbf{q}|^2 \right)^T \mathbf{h}, \end{aligned}$$

leading to the desired result. $\square$

E Appendix

Proof of Theorem 4.1. We prove it by contradiction. Let η_∞ be the infimum of (1.2). Suppose R^* is not a global solution to (1.2). Then either the infimum of (1.2) is not attainable or is attainable but R^* is not a solution. In either case, it must hold that $e(R^*) > \eta_\infty$, and we can find an $\tilde{R}(x) = \tilde{P}(x)/\tilde{q}(x) \in \mathcal{R}_{(s,t)}$ with $\tilde{q}(x_j) \neq 0$ ($x_j \in \mathcal{X}$) satisfying

$$e(\tilde{R}) := \max_{x_\ell \in \mathcal{X}} \|F(x_\ell) - \tilde{R}(x_\ell)\|_F^2 < e(R^*). \quad (\text{E.1})$$

Indeed, if the infimum of (1.2) is unattainable and $e(R^*) > \eta_\infty$, then by the definition of infimum (1.2), we can find an $\tilde{R} \in \mathcal{R}_{(s,t)}$ satisfying $\eta_\infty < e(\tilde{R}) < e(R^*)$; otherwise, if (1.2) is attainable but R^* is not a minimax approximation, then we can take $\tilde{R}$ as the rational minimax approximant of (1.2).

Let $\tilde{\mathbf{a}} = [\tilde{\mathbf{a}}_{11}^T, \dots, \tilde{\mathbf{a}}_{s1}^T, \tilde{\mathbf{a}}_{12}^T, \dots, \tilde{\mathbf{a}}_{st}^T]^T$ and $\tilde{\mathbf{b}}$ be the associated coefficients vectors of $\{\tilde{p}_{ij}(x)\}$ and $\tilde{q}(x)$ in the basis of $\{\psi_j\}$ and $\{\phi_j\}$. By item (ii) of Proposition 3.1, the matrix $H_{\mathbf{w}^*} = A_{\mathbf{w}^*} - d(\mathbf{w}^*)B_{\mathbf{w}^*}$ is positive semi-definite where $A_{\mathbf{w}^*}$ and $B_{\mathbf{w}^*}$ are defined in (3.2) and (3.3), respectively. This gives

$$\begin{aligned} 0 &\leq \begin{bmatrix} \tilde{\mathbf{a}} \\ \tilde{\mathbf{b}} \end{bmatrix}^H (A_{\mathbf{w}^*} - d(\mathbf{w}^*)B_{\mathbf{w}^*}) \begin{bmatrix} \tilde{\mathbf{a}} \\ \tilde{\mathbf{b}} \end{bmatrix} \\ &= \begin{bmatrix} \tilde{\mathbf{a}} \\ \tilde{\mathbf{b}} \end{bmatrix}^H [-\Theta, \mathbf{F}\Phi]^H \mathbf{W}_\otimes^* [-\Theta, \mathbf{F}\Phi] \begin{bmatrix} \tilde{\mathbf{a}} \\ \tilde{\mathbf{b}} \end{bmatrix} - d(\mathbf{w}^*) \begin{bmatrix} \tilde{\mathbf{a}} \\ \tilde{\mathbf{b}} \end{bmatrix}^H [0_{m \times n}, \Phi]^H \mathbf{W}^* [0_{m \times n}, \Phi] \begin{bmatrix} \tilde{\mathbf{a}} \\ \tilde{\mathbf{b}} \end{bmatrix} \\ &= \sum_{\ell=1}^m w_\ell^* \left(\sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_\ell) \tilde{q}(x_\ell) - \tilde{p}_{ij}(x_\ell)|^2 - d(\mathbf{w}^*) |\tilde{q}(x_\ell)|^2 \right) \\ &= \sum_{\ell=1}^m w_\ell^* |\tilde{q}(x_\ell)|^2 \left(\sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_\ell) - \tilde{r}_{ij}(x_\ell)|^2 - e(R^*) \right) \quad (\text{by (4.1)}) \\ &= \sum_{\ell=1}^m w_\ell^* |\tilde{q}(x_\ell)|^2 \left(\|F(x_\ell) - \tilde{R}(x_\ell)\|_F^2 - e(R^*) \right). \quad (\text{E.2}) \end{aligned}$$

This implies that there exists at least one x_ℓ so that $\|F(x_\ell) - \tilde{R}(x_\ell)\|_F^2 \geq e(R^*)$, and therefore,

$$e(\tilde{R}) = \max_{x_\ell \in \mathcal{X}} \|F(x_\ell) - \tilde{R}(x_\ell)\|_F^2 \geq e(R^*).$$

But this is a contradiction due to (E.1). This proves that the infimum in (1.2) is attainable by $R^* \in \mathcal{R}_{(s,t)}$. Thereby, we have $\eta_\infty = e(R^*)$. Since η_∞ is an upper bound of the dual problem by weak duality (2.13), $\eta_\infty = e(R^*)$ ensures strong duality (2.17). The proof is complete. $\square$

F Appendix

Proof of Theorem 5.1. The proofs for items (i) and (iii) follow that of [50, Theorem 2.4] with suitable modifications according to the block structure of (1.2).

For (i), as strong duality (2.17) holds and $(\{p_{ij}^*\}, q^*)$ are feasible (by a scaling for q^* using $\mathbf{w}^*$ to ensure $\sum_{\ell=1}^m w_\ell^* |q^*(x_\ell)|^2 = 1$) for the minimization (2.14) at $\mathbf{w} = \mathbf{w}^*$, it holds

$$\begin{aligned} \eta_\infty = d(\mathbf{w}^*) &\leq \sum_{\ell=1}^m w_\ell^* \left(\sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_\ell) q^*(x_\ell) - p_{ij}^*(x_\ell)|^2 \right) \\ &= \sum_{\ell=1}^m w_\ell^* |q^*(x_\ell)|^2 \left(\sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_\ell) - R^*(x_\ell)|^2 \right) \\ &= \sum_{\ell=1}^m w_\ell^* |q^*(x_\ell)|^2 \|F(x_\ell) - R^*(x_\ell)\|_F^2 \\ &\leq \eta_\infty \sum_{\ell=1}^m w_\ell^* |q^*(x_\ell)|^2 = \eta_\infty; \end{aligned}$$

this gives

$$w_\ell^* |q^*(x_\ell)|^2 (\eta_\infty - \|F(x_\ell) - R^*(x_\ell)\|_F^2) = 0, \quad \forall 1 \leq \ell \leq m.$$

Because $q^*(x_\ell) \neq 0$, $\forall x_\ell \in \mathcal{X}$, we have (5.2).

For item (ii), the case $e(R^*) = 0$ is trivial as $\mathcal{X}_e(R^*) = \mathcal{X}$. For the case $e(R^*) > 0$, by (5.2), we first have $|\mathcal{J}_{\mathbf{w}^*}| \leq |\mathcal{X}_e(R^*)|$ where

$$\mathcal{J}_{\mathbf{w}^*} := \{\ell : w_\ell^* \neq 0, 1 \leq \ell \leq m\}. \quad (\text{F.1})$$

Now, assume $|\mathcal{X}_e(R^*)| < \min_{1 \leq i \leq s, 1 \leq j \leq t} (n_{ij} + 2)$, and assume, without loss of generality, that $\mathcal{J}_{\mathbf{w}^*} = \{1, \dots, k\}$ with $1 \leq k \leq \min_{1 \leq i \leq s, 1 \leq j \leq t} (n_{ij} + 1)$. The later implies that for each (i, j) , we can find an interpolation polynomial $\tilde{p}_{ij}(x) \in \mathbb{P}_{n_{ij}}$ such that

$$f_{ij}(x_\ell) q^*(x_\ell) = \tilde{p}_{ij}(x_\ell), \quad \ell = 1, 2, \dots, k. \quad (\text{F.2})$$

Furthermore, by strong duality,

$$\begin{aligned} \eta_\infty = e(R^*) &= d(\mathbf{w}^*) \\ &\leq \sum_{\ell=1}^m w_\ell^* \left(\sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_\ell) q^*(x_\ell) - \tilde{p}_{ij}(x_\ell)|^2 \right) \quad (\text{as } (\{\tilde{p}_{ij}\}, q^*) \text{ may not be optimal}) \end{aligned}$$

$$\begin{aligned}
&= \sum_{\ell=1}^k w_{\ell}^* \left(\sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_{\ell})q^*(x_{\ell}) - \tilde{p}_{ij}(x_{\ell})|^2 \right) \quad (w_{\ell}^* = 0, k < \ell \leq m) \\
&= 0
\end{aligned}$$

where the last equality is due to the interpolation conditions (F.2). This contradicts against $e(R^*) > 0$, and hence the assertion (ii) follows.

To prove (iii), according to (5.2), we know for any solution $\mathbf{w}^* = [w_1^*, \dots, w_m^*]^T$ of the dual problem (2.16), $w_j^* = 0$ for any j with $x_j \notin \mathcal{X}_e(R^*)$ in (5.1). For simplicity, assume $\mathcal{X}_e(R^*) = \{x_j\}_{j=1}^s$ and $\check{\mathcal{X}} = \{x_j\}_{j=1}^{\check{s}}$ with $\check{s} \geq s$. Denote by $\check{\eta}_{\infty}$ the infimum of (5.4) and $\check{d}^* = \max_{\check{\mathbf{w}} \in \check{\mathcal{S}}} \check{d}(\check{\mathbf{w}})$. Similarly to weak duality (2.13) in Theorem 2.3, we have weak duality $\check{d}^* \leq \check{\eta}_{\infty}$; note that this weak duality holds without requiring the attainability condition of (5.4). Also, as $(\eta_{\infty}, \{p_{ij}^*\}^*, q^*)$ are feasible for (5.4), $\check{\eta}_{\infty} \leq \eta_{\infty}$; thus

$$\check{d}^* = \check{d}(\check{\mathbf{w}}^*) \leq \check{\eta}_{\infty} \leq \eta_{\infty} = d(\mathbf{w}^*) \quad (\text{F.3})$$

where $\check{\mathbf{w}}^*$ and $\mathbf{w}^*$ are solutions for the dual problems (5.5) and (2.16), respectively. On the other hand, note, by the complementary slackness, that $w_j^* = 0$ for any $j \geq s$, and hence

$$\begin{aligned}
d(\mathbf{w}^*) &= \min_{\substack{p_{ij} \in \mathbb{P}_{n_{ij}}, q \in \mathbb{P}_d \\ \sum_{\ell=1}^m w_{\ell}^* |q(x_{\ell})|^2 = 1}} \sum_{\ell=1}^m w_{\ell}^* \left(\sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_{\ell})q(x_{\ell}) - p_{ij}(x_{\ell})|^2 \right) \\
&= \min_{\substack{p_{ij} \in \mathbb{P}_{n_{ij}}, q \in \mathbb{P}_d \\ \sum_{\ell=1}^{\check{s}} w_{\ell}^* |q(x_{\ell})|^2 = 1}} \sum_{\ell=1}^{\check{s}} w_{\ell}^* \left(\sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_{\ell})q(x_{\ell}) - p_{ij}(x_{\ell})|^2 \right) = \check{d}(\mathbf{w}_{[1:\check{s}]}^*) \leq \check{d}^*,
\end{aligned}$$

where the last inequality follows because $\check{d}^*$ is the maximum of (5.5) over all $\check{\mathbf{w}} \in \check{\mathcal{S}}$. Together with (F.3), we conclude $\check{d}^* = \check{d}(\check{\mathbf{w}}^*) = \check{\eta}_{\infty} = \eta_{\infty} = d(\mathbf{w}^*)$. Finally, for any solution $\check{\mathbf{w}}^*$ of the dual (5.5), suppose $(\{\hat{p}_{ij}\}, \hat{q})$ achieve the minimum $\check{d}(\check{\mathbf{w}}^*) = \eta_{\infty}$ of (5.6). Scale $(\{p_{ij}^*\}, q^*)$ to satisfy the constraint $\sum_{\ell=1}^{\check{s}} \check{w}_{\ell}^* |q^*(x_{\ell})|^2 = 1$ (i.e., $(\{p_{ij}^*\}, q^*)$ are feasible for the minimization (5.6) with $\check{\mathbf{w}} = \check{\mathbf{w}}^*$); thus noticing

$$\begin{aligned}
\check{d}^* = \check{d}(\check{\mathbf{w}}^*) &= \sum_{\ell=1}^{\check{s}} \check{w}_{\ell}^* \left(\sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_{\ell})\hat{q}(x_{\ell}) - \hat{p}_{ij}(x_{\ell})|^2 \right) \\
&\leq \sum_{\ell=1}^{\check{s}} \check{w}_{\ell}^* \left(\sum_{i=1}^s \sum_{j=1}^t |f_{ij}(x_{\ell})q^*(x_{\ell}) - p_{ij}^*(x_{\ell})|^2 \right) \\
&= \sum_{\ell=1}^{\check{s}} \check{w}_{\ell}^* |q^*(x_{\ell})|^2 \|F(x_{\ell}) - R^*(x_{\ell})\|_F^2 \leq \eta_{\infty} \sum_{\ell=1}^{\check{s}} \check{w}_{\ell}^* |q^*(x_{\ell})|^2 = \eta_{\infty} = \check{d}^*,
\end{aligned}$$

we assert that $(\{p_{ij}^*\}, q^*)$ achieves the minimum $\check{d}(\check{\mathbf{w}}^*) = \eta_{\infty}$ of (5.6). $\square$

G Appendix

Proof of Theorem 7.2. The proof adapts [48, Theorem 5.2] with necessary adjustments. For completeness, we present the full argument here. The idea is to express and estimate the lower bound in (7.6) using the real parameter β around $\beta = 0$. For simplicity, using the notation in the proof of Theorem 7.1, we consider the nontrivial case $d(\mathbf{w}) > 0$ (the case $d(\mathbf{w}) = 0$ is trivial because $w_\ell \tau_\ell = 0, \forall 1 \leq \ell \leq m$).

Define $W(\beta) = \text{diag}(w_1(\beta), \dots, w_m(\beta))$ with

$$w_j(\beta) = \frac{w_j \tau_j^\beta}{\gamma_\beta} = \frac{w_j \|F(x_j) - R(x_j)\|_F^\beta}{\sum_{\ell=1}^m w_\ell \|F(x_\ell) - R(x_\ell)\|_F^\beta}$$

for which we have $w_j(0) = w_j$ and $w'_j(0) = w_j \log \tau_j - w_j \sum_{\ell=1}^m w_\ell \log \tau_\ell, \forall j \in \mathcal{J}_{\tilde{\mathbf{w}}}$.

In the following discussion, we can assume $\mathbf{q}(\beta)$ is continuously differentiable with respect to β around $\beta = 0$, where $\mathbf{q}(\beta) = \Phi \mathbf{b}(\beta)$ and $\mathbf{a}(\beta) \in \mathbb{C}^n, \mathbf{b}(\beta) \in \mathbb{C}^{d+1}$ are the solution to (2.12) with $W(\beta)$ (for more details, we refer to the proof of [48, Theorem 5.2]). Hence, $\mathbf{q}(\beta) = \mathbf{q}(0) + \beta \mathbf{q}'(0) + O(\beta^2)$. Since $1 \equiv (\mathbf{q}(\beta))^H W(\beta) \mathbf{q}(\beta)$, differentiating it with respect to β and setting $\beta = 0$ gives

$$\text{Re}((\mathbf{q}'(0))^H W(0) \mathbf{q}(0)) = -\frac{1}{2}(\mathbf{q}(0))^H W'(0) \mathbf{q}(0);$$

moreover, for any sufficiently small $\beta > 0$, we have

$$\begin{aligned} |(\mathbf{q}(\beta))^H W(0) \mathbf{q}(0)| &\geq |\text{Re}((\mathbf{q}(\beta))^H W(0) \mathbf{q}(0))| \\ &= |\text{Re}((\mathbf{q}(0))^H + \beta \mathbf{q}'(0)^H + O(\beta^2)) W(0) \mathbf{q}(0))| \\ &= |(\mathbf{q}(0))^H W(0) \mathbf{q}(0) + \beta \text{Re}((\mathbf{q}'(0))^H W(0) \mathbf{q}(0))| + O(\beta^2) \\ &= \left| 1 - \frac{\beta}{2}(\mathbf{q}(0))^H W'(0) \mathbf{q}(0) \right| + O(\beta^2) \\ &= 1 - \frac{\beta}{2} \left(\sum_{j \in \mathcal{J}_{\tilde{\mathbf{w}}}} |q_j|^2 w_j \log \tau_j - \sum_{j \in \mathcal{J}_{\tilde{\mathbf{w}}}} w_j \log \tau_j \right) + O(\beta^2), \end{aligned}$$

where the last equality is due to $\mathbf{q}(0) = \mathbf{q}$ and $\sum_{\ell=1}^m w_\ell |q_\ell|^2 = 1$. With this, we can write the lower bound in (7.6) as

$$\hat{\mu}(\beta) := \frac{|(\mathbf{q}(\beta))^H W(0) \mathbf{q}(0)|}{\gamma_\beta \zeta_\beta}$$

$$\geq \frac{1 - \frac{\beta}{2} \left(\sum_{j \in \mathcal{J}_{\tilde{\mathbf{w}}}} w_j |q_j|^2 \log \tau_j - \sum_{j \in \mathcal{J}_{\tilde{\mathbf{w}}}} w_j \log \tau_j \right)}{\sqrt{\left(\sum_{j \in \mathcal{J}_{\tilde{\mathbf{w}}}} w_j \tau_j^\beta \right) \left(\sum_{j \in \mathcal{J}_{\tilde{\mathbf{w}}}} w_j |q_j|^2 \tau_j^{2-\beta} \right)}} + O(\beta^2) := \mu(\beta) + O(\beta^2)$$

locally at $\beta = 0$. Note $\hat{\mu}(0) = \frac{1}{\sqrt{d(\mathbf{w}(0))}}$. For $\mu(\beta)$, by calculation, we have (with $w_j(0) = w_j$, $\mathbf{q}(0) = \mathbf{q}$)

$$\mu'(0) = \frac{1}{2\sqrt{d(\mathbf{w})}^3} \left[\sum_{j \in \mathcal{J}_{\tilde{\mathbf{w}}}} w_j |q_j|^2 \tau_j^2 \log \tau_j - \left(\sum_{j \in \mathcal{J}_{\tilde{\mathbf{w}}}} w_j |q_j|^2 \tau_j^2 \right) \left(\sum_{j \in \mathcal{J}_{\tilde{\mathbf{w}}}} w_j |q_j|^2 \log \tau_j \right) \right] \geq 0$$

where the last inequality is obtained by applying [48, Lemma 5.1]. Moreover, [48, Lemma 5.1] also says that if there is a pair (i, j) such that $i, j \in \mathcal{J}_{\tilde{\mathbf{w}}}$ and $w_i w_j (\tau_i - \tau_j) \neq 0$, then $\mu'(0) > 0$. In that case, we know that there is a $\beta_0 > 0$ such that $\mu'(\beta) > \frac{1}{2}\mu'(0)$ and $\frac{\beta}{2}\mu'(0) + O(\beta^2) > 0$ for any $\beta \in (0, \beta_0)$, implying

$$\hat{\mu}(\beta) > \hat{\mu}(0) = \frac{1}{\sqrt{d(\mathbf{w}(0))}}, \quad \forall \beta \in (0, \beta_0).$$

This shows that if $\exists(i, j)$ so that $i, j \in \mathcal{J}_{\tilde{\mathbf{w}}}$ and $w_i w_j (\tau_i - \tau_j) \neq 0$, then a sufficiently small $\beta > 0$ leads to $d(\tilde{\mathbf{w}}) = d(\mathbf{w}(\beta)) \geq d(\mathbf{w}(0)) = d(\mathbf{w})$; conversely, for a sufficiently small $\beta > 0$,

$$\begin{aligned} d(\mathbf{w}(\beta)) = d(\mathbf{w}(0)) &\implies w_i w_j (\tau_i - \tau_j) = 0, \quad \forall i, j \in \mathcal{J}_{\tilde{\mathbf{w}}} \\ &\implies \tau_j = \text{a constant } c, \quad \forall j \in \mathcal{J}_{\tilde{\mathbf{w}}}. \end{aligned} \quad (\text{G.1})$$

Obviously, $c \leq \max_{1 \leq \ell \leq m} \tau_\ell = \sqrt{e(R)}$ and also

$$d(\mathbf{w}) = \sum_{j=1}^m w_j |q_j|^2 \tau_j^2 = \sum_{j \in \mathcal{J}_{\tilde{\mathbf{w}}}} w_j |q_j|^2 \tau_j^2 = c^2 \sum_{j \in \mathcal{J}_{\tilde{\mathbf{w}}}} w_j |q_j|^2 = c^2 \sum_{j=1}^m w_j |q_j|^2 = c^2. \quad (\text{G.2})$$

Hence $\sqrt{d(\mathbf{w})} = c \leq \sqrt{e(R)}$, proving items (i) and (ii).

For item (iii), if we additionally have $c \geq \max_{\ell \notin \mathcal{J}_{\tilde{\mathbf{w}}}} \tau_\ell$, then $c = \sqrt{d(\mathbf{w})} = \sqrt{e(R)}$. According to Theorem 4.1, we conclude that $R^{(k)}(x)$ is the minimax approximant of (1.2). In this case, (7.12) can be written as $w_\ell (\tau_\ell - \sqrt{e(R)}) = 0$, $\forall 1 \leq \ell \leq m$, the complementary slackness property given in Theorem 5.1. $\square$

Declarations

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